ESSnet Big Data

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Work Package 7

Multi domains

Deliverable 7.7

The General Report for each Case Study/Domain including recommendation on legal aspects, availability, sustainability, methodology, quality and technical requirements

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1. Introduction

Among all WP7 pilots, there is a variety of data sources. We have tested several different use cases, including data sources such as business registers, web scrapped data, traffic loops, satellite data as well as Google Trends or Google Traffic. Finally, we decided to give brief description of original and reliable, sustainable data sources. However, in this document you can also find information on pilots that were tested in limited scope and not selected to present the data in the final results.

The pilots are within the following statistical domains:

- Population,
- Tourism,
- Agriculture.

All pilots have already been conducted by partner countries. For instance, two different approaches for Agriculture domain (Crop Types identifying by satellite images) have been prepared by Statistics Ireland and Statistics Poland. The case study on life satisfaction has been prepared separately by ONS UK and Statistics Poland. Moreover, our approaches have been tested by ESSNet partners (Ireland, Netherlands, Poland, Portugal, United Kingdom) with good results.

In this report you will find also information regarding data combining – what has been done in the first wave of pilots. Use cases on data combining were divided by us into two approaches:

- intra-domain (e.g., satellite data and data from in-situ surveys),
- inter-domains (e.g., agriculture and agritourism accommodation establishments).

2. Executive Summary

To be added after the chapters are completed.

- There are not so many cases in Population domain that can be implemented according to the rules of the data quality and accessibility.
- Three use cases in Tourism domain can be implemented with success – one of them can be treated as a replacement for the traditional survey – tourism accommodation establishments, the second can be treated as a supplement for traditional survey – border movement, the third – tourism visit places – can supplement the surveys already conducted.
- The implementation of the border movement pilots relies on the data availability on road traffic sensors. The data on air traffic is easily available.
- The implementation of tourism accommodation establishments is based on the availability of web data on popular accommodation portals.

- The implementation of tourism websites allows to provide information on the profile of users that visit specific tourism places (see Appendix 1.3).
- Air traffic data covering more destinations than the sample survey which gives a more stable distribution of trips for small domains.
- Agriculture domain has one successful project – Crop types identification that is compliant with in-situ survey on crop types. It can be used to replace the surveys on crop types in agricultural region. It is based on satellite data.
- Population domain in Big Data is strongly related to the social statistics – we are estimating life satisfaction according to the human sourced information – comments, posts and blogs in a specific term.
- The pilots on Life satisfaction within Population domain have been shared on Github repositories and are easy to implement by other countries – they are using Machine Learning algorithms to identify the sentiment.
- Pilot on Tourism accommodation establishments can be implemented with the use of web scraping methods.

WP7 prepared and tested 6 intra-domain pilots in three different domains (3 Population, 2 Tourism, 1 Agriculture with 2 pilots implemented by different approaches, first by Statistics Poland and the second by CSO Ireland). For data combining we have two different approaches – intra-domain data combining (combining of different data sources within one domain – e.g., survey data and web data) and inter-domain data combining (combining data sources from two or more domains, e.g., agriculture-tourism).

3. Population

3.1. Introduction and overview

The goal of Population use cases was to show the structure of population in different regions according to the specific facts – e.g., public opinion on a topic (in the pilot Brexit) and life satisfaction in different regions.

Therefore, three use cases have been conducted. One of them was to identify the scale of the depression in different countries based on the Google Trends. The second was to find social mood according to public events or facts (e.g., Brexit). The goal of the third use case was to identify life satisfaction in population according to their comments/posts/tweets.

3.2. Intra-domain analysis

The aim of this section is to provide an overview of the use cases related to population domain by three perspectives:

- possible data sources,
- successful use cases conducted,
- overview of the results.

Possible data sources

The goal of the intra-domain analysis was to identify possible data sources and conduct pilot use cases within population domain. Main data sources identified to conduct the pilot on life satisfaction were Twitter and Facebook. Below you can find the most important information regarding these data sources.

> Twitter and Facebook users differ from the general population in terms of demographics, political attitudes and political behavior. We find that Twitter and Facebook users differ substantially from the general population on many politically relevant dimensions including vote choice, turnout, age, gender, education and region.
>
> (*Twitter and Facebook are not representative of the General Population: Political Attitudes and Demographics of Social Media users* https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2791625)

Depending on the data source, several different attributes can be identified. In the table below, you can find the possible attributes by data sources.

Table 1. Possible attributes of population by the type of data sources

	Twitter	**Facebook[1]**	**Instagram**	**LinkedIn[1]**	**Comments - registered users[1,2]**
Gender	✓	✓	✓		✓
Age	✓	✓	✓	✓	✓
Location	✓	✓	✓	✓	✓
Ethnic Origin	✓	✓			
Children	✓	✓			
Marital status		✓		✓	
Interests	✓	✓	✓		
Occupation		✓		✓	
Education		✓		✓	

1) Access to user's account data is restricted.
2) A lot type of sources like blogs, news websites etc. Most common data.

✓ Free access by API
✓ Paid or restricted access
✓ Indirect access

Not all of the attributes can be easily extracted from the data. For example, gender by Twitter can be identified with machine learning algorithms as it is not necessary to share this information in the user profile. Data on population from different social media portals can be extracted by the API (Application Programming Interface) or indirect.

Table 2. Comparison of different data sources by possible attributes of population

Characteristic	Source	Based on	Metadata
Twitter			
Gender	Indirect	Profile's colours; User names; User tweets; Facial evaluation	Language Independent Gender Classification on Twitter https://www.cs.uic.edu/~buy/Asonam2013.pdf
Age	Indirect	User tweets; User followers, friends and activity;	Predicting age groups of Twitter users based on language and metadata features, https://doi.org/10.1371/journal.pone.0183537
Place of living/ Urban and village inhabitant	API	User locations; Tweet geographical coordinates	location; coordinates
User activity by country	API	User locations; Tweet geographical coordinates; timezone	location; coordinates; time_zone
Having children	Indirect	Hashtags; subscribed profiles; groups memberships	hashtags; GET friends/list GET lists/memberships
Interests	Indirect	Subscribed profiles; groups memberships	GET friends/list GET lists/memberships
Gender, Age, Ethnic origin	Indirect	Hashtags used by "groups"	*#greysanatomy vs. #yankees: Demographics and Hashtag Use on Twitter* https://arxiv.org/abs/1603.01973
Facebook			
Gender	API or indirect	Gender information Facial evaluation.	gender; picture
Age	API	Age information	age_range; birthday
Place of living/ Urban and village inhabitant	API	Address and language information	adress; language; timezone
Ethnic origin	API	Person's hometown	hometown Private Traits and Attributes are Predictable from Digital Records of Human Behavior, http://doi.org/10.1073/pnas.1218772110

Education	API	Names of schools graduated (including achieved grade and year)	education;
Children, their gender and age	API	Family relationships	family;
Marital status	API	Family relationships	relationship_status; family
Occupation	API	Workplace, work experience	work
Interests	API	Subscribed profiles;	subscribeto; likes; movies; music; books; favourite_athlets; favourite_teams political
Instagram			
Gender Age	Indirect	User information	GET/users/self GET /users/user-id
Interests	Indirect	Subscribed profiles	GET /users/self/follows GET /users/self/media/liked GET /tags/tag-name/media/recent
Place of living/ Urban and village inhabitant	API	Address information	GET /locations/location-id GET /locations/search

Often, the profiles are not public, the data is incomplete or not available, sometimes untrue.

It is much easier to access Twitter data than Facebook data. There are little restrictions to access and process Twitter data comparing to Facebook data. However ONS UK has conducted a use case using also data from Facebook and Guardian website. The pilot developed for the WP7 using Facebook API can be found here:

https://github.com/ONSBigData/ESSnet-WP7-UK-Case-Study

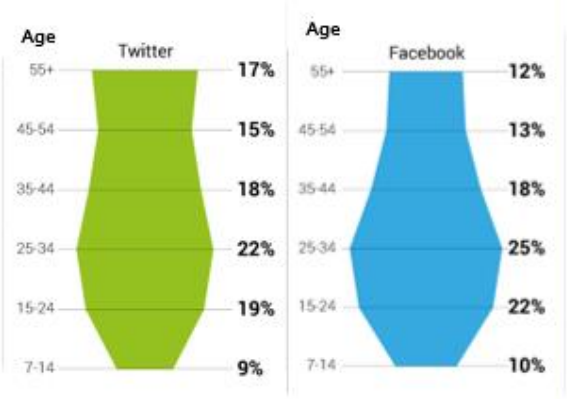
This pilot is very promising as Facebook is much more popular in many countries and more often used by its owners than Twitter. Short comparison of the popularity of Twitter and Facebook social media is presented below.

➢ COVERAGE OF TWITTER AND FACEBOOK

https://www.tweetping.net/#/ https://www.facebook.com/notes/facebookengineering/visualizingfriendships/469716398919

➢ USERS IN POLAND

3,3 mln **20,1 mln**



*source: *Gemius; June 2015*

➢ CHARACTERISTICS OF COMMENTS AND OPTIONS

Twitter

- Short comments currently limited to 280 characters (photos, links, username also include this limit);
- Hasztags # are the keys - they make it easier to access information;
- Retweets- forwarding information;
- Option to like, comment;
- Comments are informative. Often comments refer to a given field, users usually do not comment on their personal lives;
- People with similar interests are observed.

Facebook

- Sharing, transfer;
- Friends are usually people which we know;
- Emoticon options: like, super, haha, wow, I'm sorry, wrr;
- Comments on Facebook are more personal.

„If something happens, then on Twitter it appears on average within 20 minutes. On Facebook after 4 hours... "

➢ LINKAGE:

- Analysis of the phenomenon according to the regional key;
- Time of year;
- Time of day;
- Date;
- The mood depends on the socio-political situation - reaction to events;
- Gender.

➢ OBSTACLES:

- Option to connect the Twitter and Facebook accounts - duplicating comments wil affect the results of research difficulties in verifying;
- Retweet, sharing.

Obtaining information on the population of the units in a big data source is important for reliable estimates. However, this is not an easy task as many of the commonly used identifiers for statistics, so-called background characteristics, are absent in many big data sources. Extracting features is a way to deal with that challenge. However in Twitter, it has been demonstrated that gender can be reliably derived from a combination of features[1].

Another issue is that big data can be created by mixed populations of units. For example, on social media persons, companies and other types of organizations are active. It might even be that units from other countries are included. For messages studied in the Netherlands for example, the question is if these are created by accounts of Dutch units or not. Statistics Netherlands work on a large Twitter dataset used by Dutch Universities revealed that the population producing those messages in 2015 contained 6.2 million unique Twitter user IDs[2]. A very computational efficient process was developed to identify Dutch accounts with an overall recall of 0.93 (Figure below). In the end, 1.25 million Twitter user IDs were identified as Dutch, demonstrating that a large part of the population was not Dutch or had an unclear location. To discern between Dutch and Flemish messages a Machine Learning approach was developed (bottom left of figure 1).

Another example of mixed populations is that of accounts used by persons or companies. This was also studied for Twitter. Based on a whole set of features, machine learning methods were developed to discern between persons and non-persons accounts and between companies and other non-persons accounts. In this way, persons could be discerned with an accuracy of 0.98. From the non-persons part, companies could be identified with an accuracy of 0.89.

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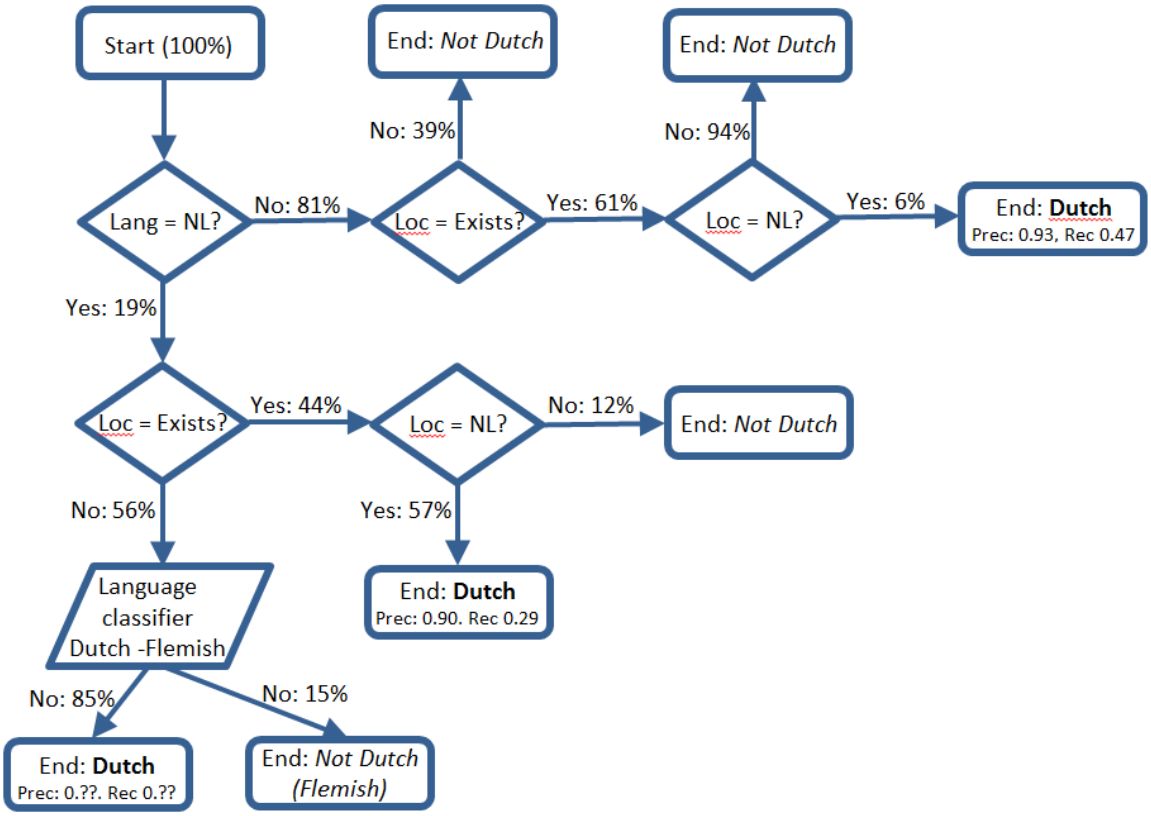
[1] Daas, P.J.H., Burger, J., Quan, L., ten Bosch, O., Puts, M. (2016) Profiling of Twitter Users: a big data selectivity study. *Discussion paper 201606*, Statistics Netherlands, The Hague/Heerlen.

[2] Daas, P., Puts, M., Renssen, R. (2017) On Big Data based Statistical Inference. Abstract and poster for the 3rd UCL Workshop on the Theory of Big Data, June 26th-28th, London, UK.

Figure 1. Overall classification scheme to identify Dutch Twitter accounts.



Successful pilot - PL/PT/UK Life Satisfaction

The most promising use case is on Life Satisfaction based on Twitter data. The goal of this use case is to identify the sentiment according to EU-SILC attributes. The source code for this pilot has been shared on Github repository with the following address:

https://github.com/jmaslankowski/WP7-Population-Life-Satisfaction

The implementation of this use case can be divided into three steps:

- preparing a training dataset,
- testing the training dataset and improving it,
- estimating life satisfaction with Twitter streaming data.

The first step is to prepare a training dataset. It is necessary to use a registered Twitter account or register a new account and get API keys to collect information from Twitter by different keywords and countries of tweets. We can use geolocation functionality to limit tweets for a specific region. After collecting the

list of tweets, manual work of tweets identification according to EU-SILC classification of Life Satisfaction is needed.

Based on collected CSV files with tweets, each tweet is identified according to the European Social Survey, Social Cohesion Survey and EU Statistics on Income and Living Conditions (EU-SILC):

- happy
- neutral
- calm
- upset
- depressed
- discouraged
- indeterminate

The list as the final results can look like this:

- 1;happy;Evening walk :) Something pleasant for the rest of the day :D #walk #winter #rest;1
- 2;neutral;#walk in the winter #rest;2
- 3;neutral;Have a rest in the city. #rest #weekend;2

The goal of the second step is to assess the training dataset accuracy.

precision recall f1-score support
1 0.90 1.00 1.00 2000
2 0.77 1.00 0.80 2000
4 0.90 1.00 1.00 2000
5 0.77 1.00 0.80 2000
7 0.90 0.33 0.50 2000
avg / total 0.87 0.80 0.77 10

Precision is a rate of positive predictive values and recall is a rate of relevant instances retrieved over the total amount of relevant instances. F1-score is represented by the equation of precision and recall. Support is the number of trained observations. We accept the precision of more than 80%. If the precision is lower (e.g., item 2 and 5), we have to improve the training dataset.

The goal of the third step is to collect the tweets from the Twitter streaming data for the specific region and provide results.

Results obtained – input for experimental statistics

As the most promising pilot in Population domain was a life satisfaction of social media users based on Twitter by countries, we decided to disseminate the results. According to EU-SILC classification, the following cafeteria has been used: 1.happy, 2.neutral, 3.calm, 4.upset, 5.depressed and 6.discouraged.

The following countries used the same approach proposed by Statistics Poland: Poland, Portugal and United Kingdom. However, data scientists from INE Portugal and ONS UK decided not to use the same classification as prepared by Statistics Poland. They excluded from original classification the neutral item. Also ONS UK decided to merge depressed and discouraged into one item. The available data are indicated in green in the table below.

Table 3. Life satisfaction of Twitter users by country

Country	happy	neutral	calm	upset	depressed	discouraged
Poland						
Portugal		n/a				
United Kingdom		n/a			merged	

The data from this pilot are going to be shared in ESSNet Big Data experimental statistics section on Wiki webpage. The results for experimental statistics are planning to be presented one time for three countries mentioned in the table above, with respect to the dates of ESSNet Big Data SGA-1.

3.3. Main findings

Preparing the use cases in Population and conducting pilots provide the following conclusions:

- there is still little information available on the Internet to provide reliable data in population domain,
- the structure of population to be analyzed is strictly related to social media users or website users,
- the only way to conduct pilots of population is to select specific group of users (e.g., Twitter users) and make analysis of its structure by regions/gender and different aspects of life satisfaction,
- although the data source is not representative for the whole population, the results are very promising in terms of accuracy of machine learning algorithms,
- with current data sources, we can support social surveys, such as EU-SILC to provide information on selected life satisfaction among people – therefore a subpopulation of social media users can be an additional information to supplement the data from the survey,

- the value added of the pilots in population domain is that we can do it every day to analyze the causes of changing moods/opinions/life satisfaction,
- the use cases on population/life satisfaction are easily replicable to other countries as the source code is publicly available on the Internet – the most time-consuming work is to prepare a training dataset,
- EU-SILC results are different than the ones received by social media sentiment analysis, probably due to the coverage and representativeness issues,
- negative things appear faster in social media than positive (e.g., bribery), also the fake news spread very fast,
- people are more willing to comment on news when they are angry.

We see the following future perspectives for Population domain:

- Social media is very useful to find the sentiment of the population – this use case should be developed by rather to provide information on whether people are angry or not, than to estimate the number of population.
- We should not concentrate just of Facebook or Twitter because they may not exist in the future.
- EU-SILC classification must be revised because in social media it is difficult to prepare a good training dataset to differentiate all life satisfaction attributes (happy, neutral, calm, upset, depressed, discouraged) and for the social media sentiment analysis we should try to find the best to identify the sentiment in reliable way.

4. Tourism

4.1. Introduction and overview

As a result of the inclusion of Poland in the Schengen Information System, on 21 December 2007 the Border Guard discontinued the registration of movement of persons and means of transport at the borders with the countries of the European Union. New regulations on customs clearances resulted in the loss of data on border traffic at the Polish borders with EU member states including land, air and maritime border.

Therefore, methodological work has been undertaken in order to develop methods for collecting the missing data on border traffic. Under the Law on official statistics, since 2014 a sample survey titled “Trips made by non-residents to Poland”. The movement of vehicles and persons at the Polish border with the countries of the European Union has been conducted. The aim of this study is to assess, on a quarterly basis, the number of trips to Poland made by foreign tourists and same-day visitors, who cross the Polish border with respect to country of origin and

destination. Information regarding border traffic is developed based on the results of the aforementioned survey and secure the information needs of official statistics in the field of tourism, balance of payments, national accounts and cross-border areas.

Significant part of survey “Trips made by non-residents to Poland” is estimation of border traffic intensity with manual counting of vehicles, including plates and type of vehicles and with data obtained from Civil Aviation Authority of Poland. Border traffic is a base for weights calculations. In fact, all of the further results depends on that.

Manual counting is very time-consuming while it requires very basic activities. We agreed that all of information gathered by interviewers may be collected without their participation: data on number of vehicles with respect to type may be gathered by road sensor weighting vehicles, country ID from plates can be identify by cameras.

Regarding to air traffic, air transportation is much more controlled than land traffic. It gives an opportunity to use free-of-charge reliable data e.g. on flight schedules to improve air traffic statistics even in costless way.

Therefore, we focused on available data from traffic loops to enhance land border traffic statistics as well as on online sources on air traffic to enhance air traffic and tourism statistics.

Domestic rail transportation was also considered as a source of domestic tourism statistic but problem of domestic transportation is much more complex. Thus we point it as a matter of further concern.

4.2. Intra-domain analysis

Road traffic

We investigated available data from traffic loops in Poland as well as in and neighbouring countries from Schengen area.

The data owners in original language are:

- Generalna Dyrekcja Dróg Krajowych i Autostrad for Poland (GDDKiA)
- Bundesanstalt für Straßenwesen for Germany (BASt)
- Ředitelství Silnic a Dálnic for Czech Republic (RSD)
- Národná diaľničná spoločnosť for Slovakia (NDS)
- Kelių ir transporto tyrimo institutas for Lithuania (KTTI)

We successfully contacted GDDKiA, BASt, KTTI and NDS. At this moment we started cooperating GDDKiA, BASt and KTTI. Data is provided regularly but with very different timing. Data

from GDDKiA and BASt is obtained with insufficient timing to employ new data into data processing for purposes of survey but it can be utilize as an additional variable to estimate border traffic. Data from KTTI is obtained with sufficient timing to use in border traffic survey conducted by our Statistical Office.

BASt and KTTI obtain data only from traffic sensors because their sensor net is really dense. Obtained data is available from 2003 for BASt and from 2007 for KTTI.

GDDKiA conducts two types of measurements called Continuous Traffic Measurement and General Traffic Measurement. Continuous Traffic Measurement is based on over 100 traffic sensors which collect data 24/7 mostly. At this moment the data is available for period 2012-2015. General Traffic Measurement includes manual, semiautomatic and automatic counting of vehicles. Result of that survey are available each 5 years (2005, 2010, 2015) for 1952 points on motorways and 2923 point on national roads. Manual and semimanual counting is a sample survey. The measurement is conducted for 9 or 5 days per year according to traffic intensity observed in previous survey. Annual average daily traffic (AADT) is calculated basing on weighting average of measurements.

Data analysis

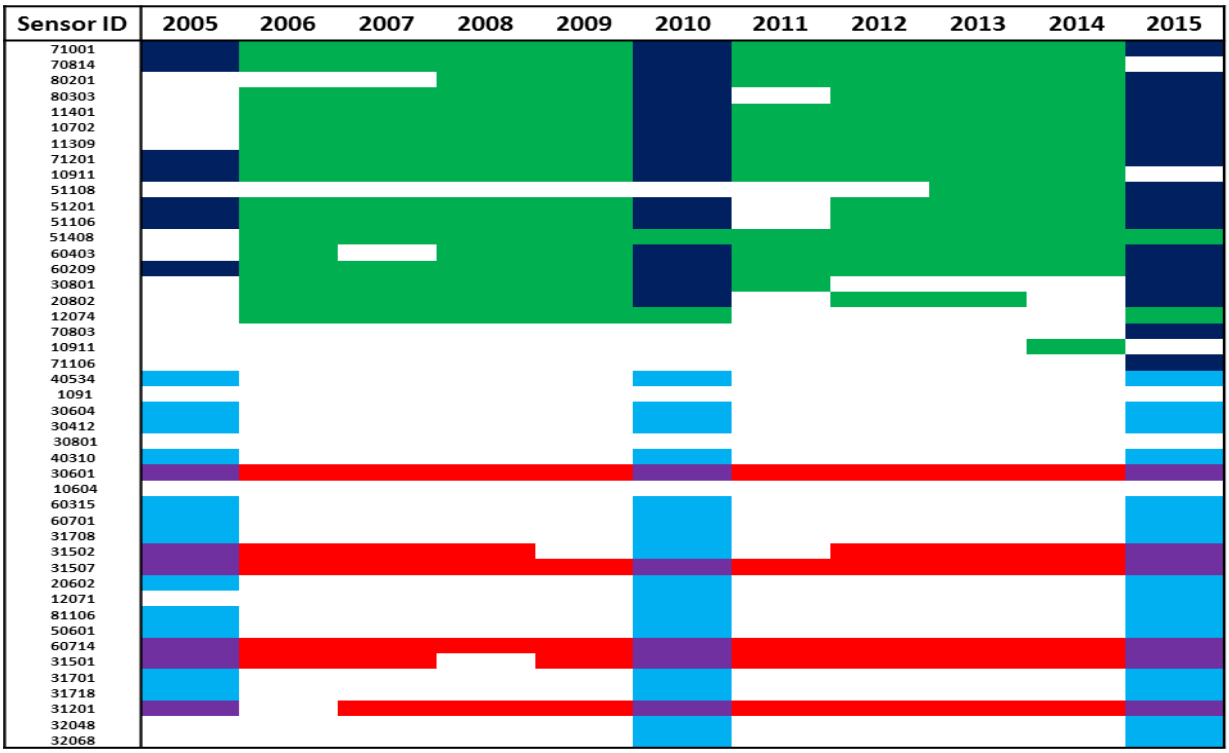
Since the database from General Traffic Measurement consisted of almost 5 thousand sensors we selected those near border by filtering those with route ending on border of state. On the one hand it caused that those sensor where well-located usually. On the other hand we omitted sensors which were placed further from the border and could be much less accurate but also they could refer to border crossings. Therefore, in a final result we expected to obtain underestimation of border traffic but without substantial bias. Thus, trends from partially estimated of border traffic should follow true border traffic.

During data collection and analysis we observed the following problems:

1) Large amount of data but only at several time points (general measurement),
2) High frequency data but in small amount (continuous measurement),
3) Irregular data gaps,
4) Recording of internal traffic (bias of result),
5) Changes of road network.

To visualize issues 1)-3) we picked part of database and with colour we marked where the data is available as below.

Figure 1 Data availability for border traffic estimation – sample of database



Source: traffic loops data from Poland and neighbouring EU's countries

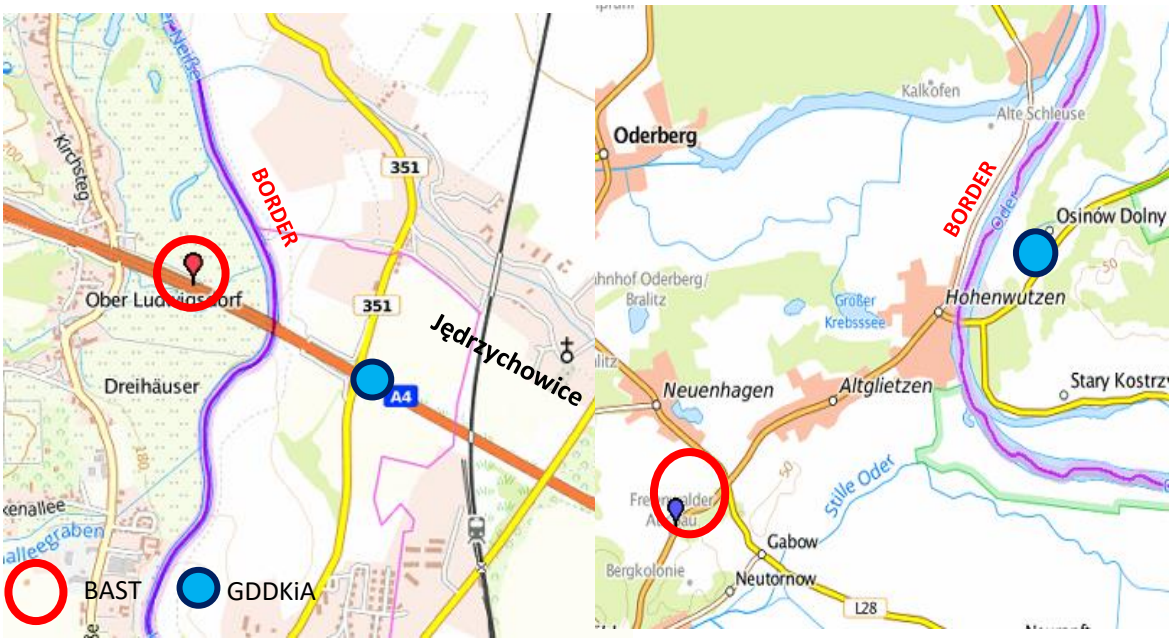
Green colour represents data from Continuous Traffic Measurement. It is high frequency data but in small amount. Light blue colour represents data from general measurement. It is large amount of data but only at several time points. Red colour represents data from BASt. With navy and purple colour there are marked points where two sources of data are available. It can be observed that data gaps are very irregular. Gaps at the beginning of time series seem to mean that the sensor have not been run yet, gaps at the end mean that the sensor has been moved. Surprisingly gaps occur even in the middle of time series. There are many traffic sensor with data available only for three years. None of year can be treated as the most "complete" year. Even in 2015 there are missing values for some time series which had values in period 2005-2014.

Summing it up:

- there is no year that can be used as a benchmark,
- gaps are so irregular that no time-series model can be used.

Since we obtained data from Polish authority and foreign authority which serves as a mirror statistics we could compare data for similar locations of traffic sensors. For some locations difference between Polish and foreign sensor was too big and regular to be considered as a random. Figure below presents two situations: good and poor location of traffic sensors along Polish-German border. In the left map sensor are located close to border. There are no roads between sensor and the border. Therefore, traffic sensor records only border traffic. The difference between sensor measurements for both sides of the border is random. In the right map German sensor is located far from the border. There are roads crossing the route between sensor and the border. Thus sensor records domestic and border traffic hence the measurement is biased.

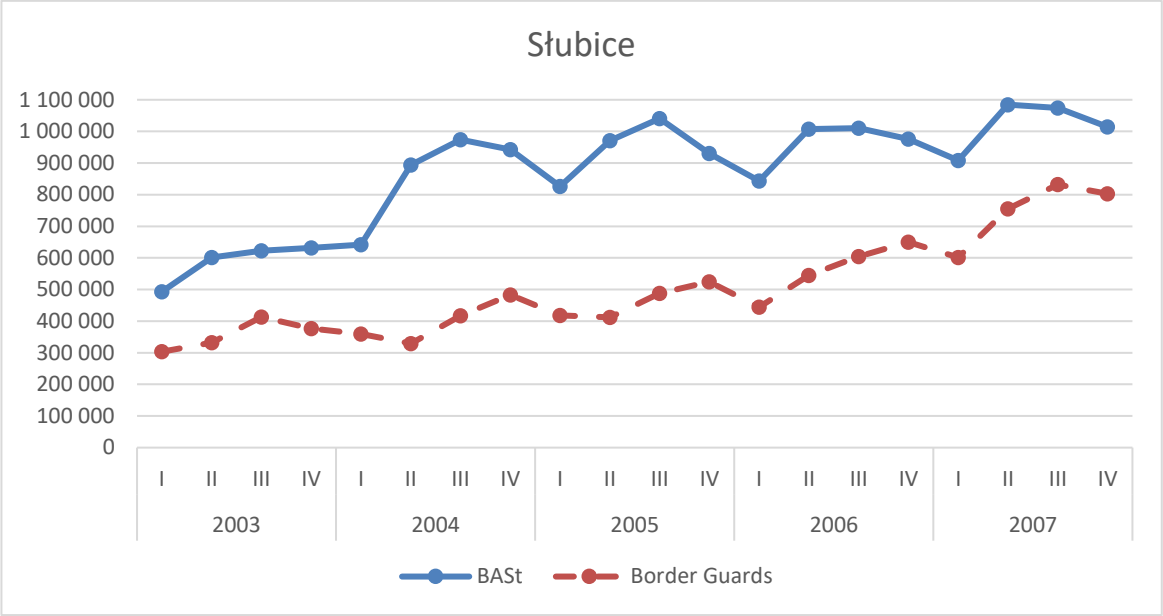
Figure 2 Traffic sensor location



Source: https://www.bast.de/BASt_2017/EN/Home/home_node.html

The question was posed – is it possible to measure the bias somehow? The only way was to find an unbiased source available for the same period of time and relevant locations as traffic sensors. We checked period before 2007 when Border Guards provided detailed data on border traffic e.g. for Polish-German border. The data from Border Guards was compared with German traffic sensor data. Next figure shows magnitude of bias for one of the border crossing.

Figure 3 Traffic measurement bias



Source: Polish Border Guards data and BASt data

For several years number of vehicles measured near the border in Słubice with traffic sensor was twice more than number of vehicles crossing the border reported by Border Guards. The difference was stable and one-directional.

We cannot use BigData directly if data is biased because we cannot use statistical method to combine biased and unbiased sources under assumption of random difference between sources. It made us wonder

- how to reduce biasedness if we have unbiased source of data but only for the past?
- how to deal with situations when we expect biasedness but there is no unbiased source in past?

We found very base solution for the first question posed. Unbiased-to-biased data ratio was calculated. Simple extrapolation of series of that coefficient was used to make the data to be unbiased. We have just multiplied biased data by forecasted coefficient. It caused there was a better comparability of data between sources. We are aware that long-term forecasts are inaccurate but it was better to apply any reasonable method than none.

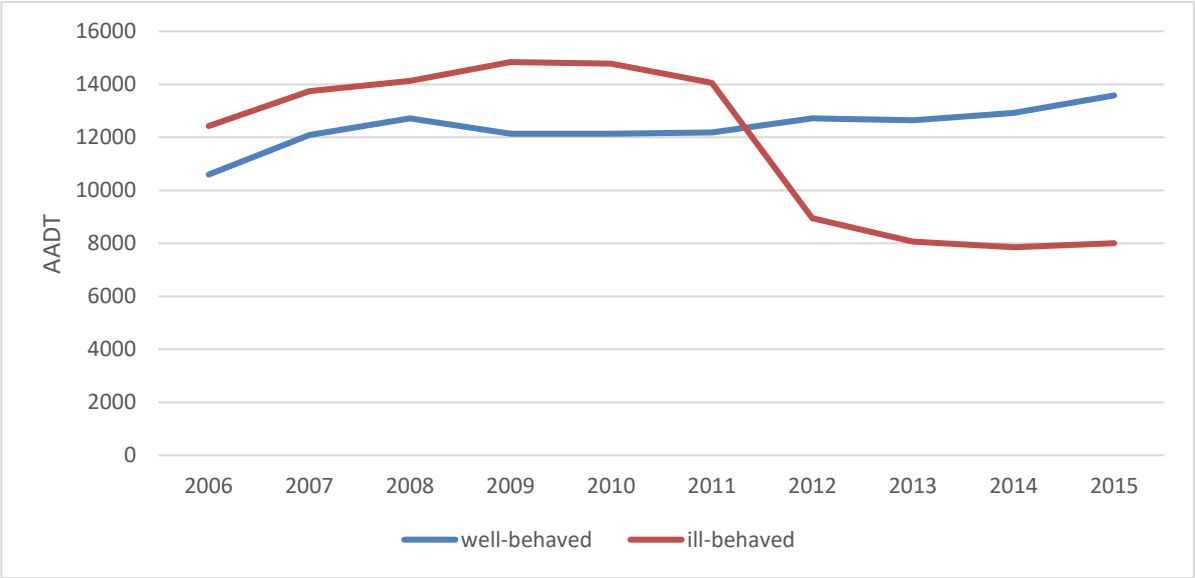
The second question at this moment remains without answer.

Summing it up:

- we cannot use BigData directly if data is biased because we cannot use statistical method to combine biased and unbiased sources under assumption of random difference between sources
- methods for reducing bias need to be developed in a case if unbiased source of data is available and it is not

One thing that was very hard to capture are changes in road network. Basic temporal analysis revealed that most of the time series exhibit nice regularities. Simply, checking R-squared of linear trend model we obtained that more than 47% of time series of relevant length of Continuous Traffic Measurement has R-squared greater than 0,7. That allows us to produce nice forecasts for these time series. On the other hand we discovered that some time series have huge fall (level shift) probably connected with changes in traffic network. Example of well-behaved and ill-behaved time series are presented below.

Figure 4 Traffic intensity at well-behaved and ill-behaved point, over the period 2006-2015



Source: GDDKiA data

We developed road traffic model which includes several algorithms basing on the concept of cross-entropy. Data processing is descripted in Appendix 1.

Results for road traffic

We used traffic sensor data available for Poland, Germany, Lithuania and Slovakia to estimate road traffic with Schengen area for the period 2005-2015 which is unavailable since 2008.

At this moment we work on partial replacing of sample survey data on road traffic by traffic sensor data.

Tables below present monthly data on light and heavy vehicles traffic in 2015 with respect to borders.

Table 4. Light vehicles (cars, minibuses and motorcycles)

	Czech Republic	Lithuania	Germany	Slovakia
January	1 724 596	133 529	2 806 735	464 488
February	1 936 668	128 860	3 113 467	526 317
March	1 932 872	133 361	3 161 557	513 854
April	2 017 946	144 545	3 295 351	527 700
May	2 313 822	169 537	3 678 351	608 747
June	2 231 358	169 472	3 654 062	601 474
July	2 502 593	203 610	4 066 808	697 140
August	2 637 689	220 525	4 325 475	744 201
September	2 235 375	168 530	3 694 648	609 959
October	1 906 322	136 343	3 253 412	512 311
November	1 281 670	90 751	2 255 733	333 466
December	842 811	58 976	1 472 634	211 708

Table 5. Heavy vehicles (trucks, non-passenger vehicles over 3,5 tone)

	Czech Republic	Lithuania	Germany	Slovakia
January	417 747	134 285	901 069	75 884
February	458 710	170 419	1 030 315	77 282
March	507 395	171 414	1 062 723	100 230
April	460 530	164 794	998 715	96 036
May	501 821	181 523	1 205 794	99 288
June	551 765	177 359	1 170 935	96 302
July	540 619	175 784	1 203 855	68 270

August	541 909	175 849	1 182 562	57 695
September	545 583	178 608	1 125 829	89 085
October	546 162	170 178	997 998	100 805
November	450 717	125 962	746 297	101 720
December	286 112	82 259	481 681	71 799

Table 6. Buses

	Czech Republic	Lithuania	Germany	Slovakia
January	12 629	1 142	19 732	5 066
February	11 052	1 061	18 713	5 486
March	13 003	1 410	19 237	6 855
April	15 736	1 957	20 270	7 567
May	22 812	3 199	25 640	10 398
June	22 911	3 382	28 710	12 451
July	22 979	3 289	33 047	10 664
August	23 125	3 348	31 835	8 735
September	22 221	2 720	28 292	10 460
October	18 814	1 914	23 290	12 387
November	10 952	868	13 491	6 065
December	6 560	481	9 773	3 891

Source: Own elaboration

To check the consistency of BigData results with other sources results were compared with sample survey on border traffic conducted by Statistical Office in Rzeszów. The crucial statistics used from sample survey were number of people crossing the border and average number of people in vehicles. The average from sample survey was compared with average from BigData. For every border the average from BigData was higher than from sample survey. Using that fact we were able to adjust number of vehicles to obtain the average number of people in vehicles. In that way we got expected number of vehicles and compared it to figures from BigData only. Next table

presents details and coverage – what amount of border traffic was estimated from traffic sensors only.

Table 7. Accuracy of border traffic estimations from BigData

	Czech Republic	Lithuania	Germany	Slovakia
Vehicles from BigData	29 575 585	3 691 245	51 158 035	7 485 785
People / Vehicles from BigData	1,88	2,09	2,13	4,19
People / Vehicles from survey	1,72	1,34	1,60	2,04
Expected vehicles from BigData	32 327 812	5 725 586	67 972 905	15 411 419
Coverage	91%	64%	75%	49%

Source: Own elaboration

Summing it up:

- well-located traffic loops can be used for estimation covering 49%-91% of road border traffic
- coverage can be increased by including other traffic loops but with greater bias
- at this moment only one data owner provides data with relevant timing to use it in border traffic estimation. The other own provides data that can be used as additional variable in the time series model

Air traffic

Tourism makes a major contribution to the global economy. According to "CITY TRAVEL & TOURISM IMPACT 2017" in 2016 inbound spend on city travel and tourism for Warsaw amounted 4.7 bln of euros. It is 4.9% of GDP of Warsaw.

Aviation plays a central role in supporting tourism. Over 54% of international tourists now travel by air. Moreover, air traffic is well-organized and controlled in comparison with road traffic. It gives an opportunity to utilize load of information that is outside official statistics to capture the increasing role of aviation in tourism and tourism itself needs.

Since lots of data on air traffic are already available online we focused on online resources. There are many websites with global flight tracking service that provide with real-time information about thousands of aircraft around the world. Each day number of flights tracked may reach 170 ths. At the beginning of our analysis we focused on web scraping of flight movement webpages.

Data were collected day-by-day with web scraping tool. Data currently obtained concerns 29 airports: 15 Polish airports and 14 hub airports (selected according to traffic intensity from Polish airports). They consist of:

- Origin and destination airport,
- IATA and ICAO codes,
- Type of aircraft,
- Date of arrival and departure.

To retrieve origin and destination country we use base of IATA, ICAO and FAA codes. In the base there is information on airport name, city, country, IATA, ICAO and FAA codes for over 10 ths. of airports.

With web scraping tool we collected data on technical information on aircrafts. They contained information on:

- Type of aircraft,
- Airline
- ID
- Total number of seats and seats in each class.

We also gathered administrative data from Civil Aviation Authority of Poland on number of passenger from Polish airports in domestic and foreign traffic. Foreign traffic data cover only direct flights from Poland which gives much less countries than from sample survey that is conducted by Statistical Office in Rzeszów. Therefore that data was utilized as a benchmark.

Our estimations would not be possible without using some crucial statistics from sample survey aforementioned. On the other hand statistics from the survey were used in limited way.

Data analysis

First step was to investigate direct flights only from Poland - to 198 airports in 54 countries. 54 countries it is much less than from sample survey which covers around 80 countries. Therefore we decided to include also flights from hub airports. We selected crucial hub airports in e.g. Germany, Netherlands, Austria, United Kingdom basing on number of direct flights. We obtained data on flight to 484 airports in 145 countries by combining air traffic data and IATA/ICAO code base.

The following air traffic data issues were identified:

- Duplicates,
- Origin=destination flights,
- Coding problems (lack of codes, different country name in IATA/ICAO code base and in air traffic data),
- IATA, ICAO and FAA codes base needs updates. During one quarter we needed to add information on few new airports to base.

Applying Big Data sources to trips statistics and air traffic we discovered that Big Data solely is still not enough. We encountered lot of issues that we were not able to tackle with Big Data only e.g.:

1) What is a share of Polish residents in total air traffic?
2) How often travellers use certain airport as a hub airports?
3) In what way we can switch from flight analysis into trips/passenger analysis?
4) Can we improve or benchmark estimated number of trips to hub airports/countries?
5) How to deal with incomplete IATA/ICAO in database?

With number of flights it was not possible to retrieve number passenger easily due to different size of aircrafts. Capacity of aircraft is different for each airline even for the same type of aircraft, thus distribution of flights is not the same as distribution of passengers Therefore, web scraping of plane seats webpages was included. We obtained information on aircraft type, airline, total number of seats and in each class.

Administrative data from Civil Aviation Authority of Poland on number of passenger from Polish airports in domestic and foreign traffic allowed us to adjust number of trips to hub airports and some but not all other countries. Therefore number of trips to the first-step countries was estimated very well.

From sample survey we estimated some crucial indicators which are not available in others source:

- percent of Polish resident in total international air traffic,
- percent of Polish resident using hub airports with respect to crucial hub airports

For instance, approximately 20% of passengers in air traffic travelling from Poland to Germany will continue their trip to other countries.

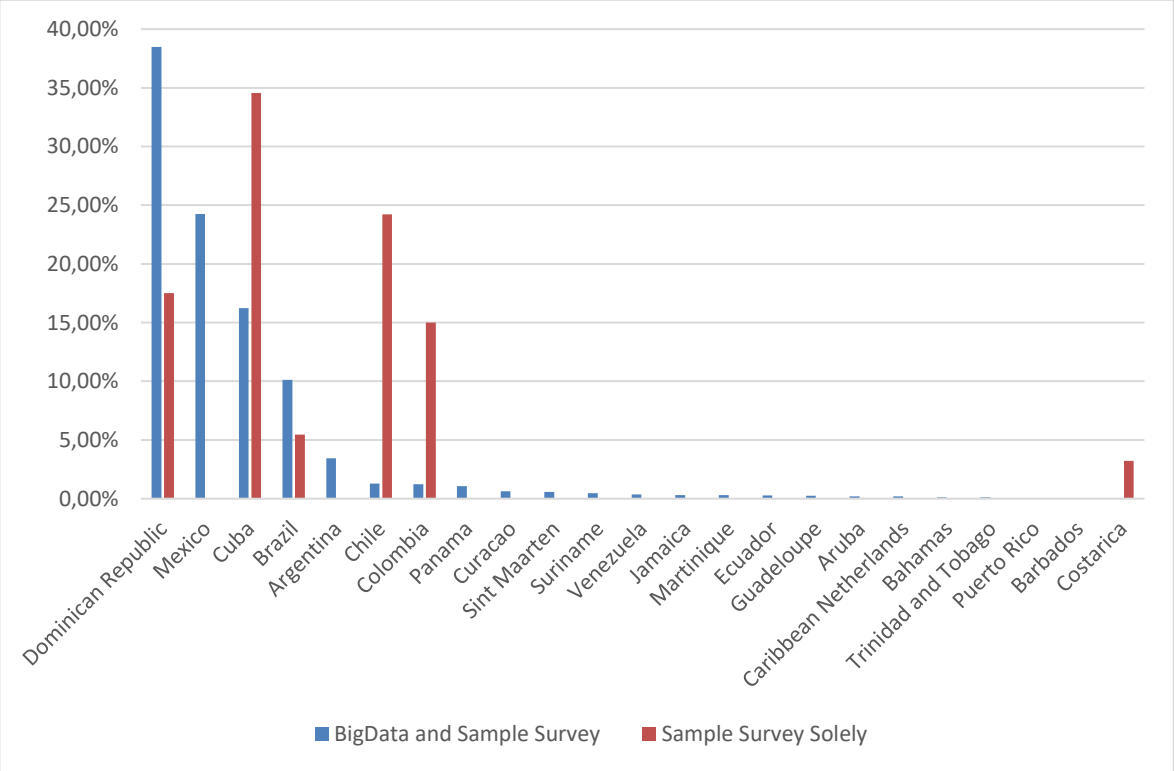
We developed air traffic model which includes several data sources linked together. Data linkage is complex but processing is quite easy. Details are described in Appendix 1.

Results for air traffic

We ran air traffic model for the third quarter of 2017. We obtained estimates for 145 countries while sample survey covers up to 80 countries.

One of the issues we examined is a distribution of trips for small domains such trips to distant countries e.g. in Latin America. It turned out that combination of Big Data and sample survey gives smoother distribution of trips. With sample survey solely we reached only 6 destinations in Latin America and the distribution seems to be distorted and unreliable. At the same time with Big Data we reached 22 destinations out of around 30 possible destinations. Moreover, Big Data ensures much less randomness in distribution then sample survey.

Chart. Distribution of trips basing on sample survey solely and with Big Data.



Source: Own elaboration

Issue to concern is the distribution of flights reflects preferences of all nations – not Polish directly. Therefore, obtained results may be more European than Polish. Therefore, in future we may examine how combine slightly biased but stable distribution received from Big Data and unbiased but unstable distribution received from sample survey. Table below presents our results.

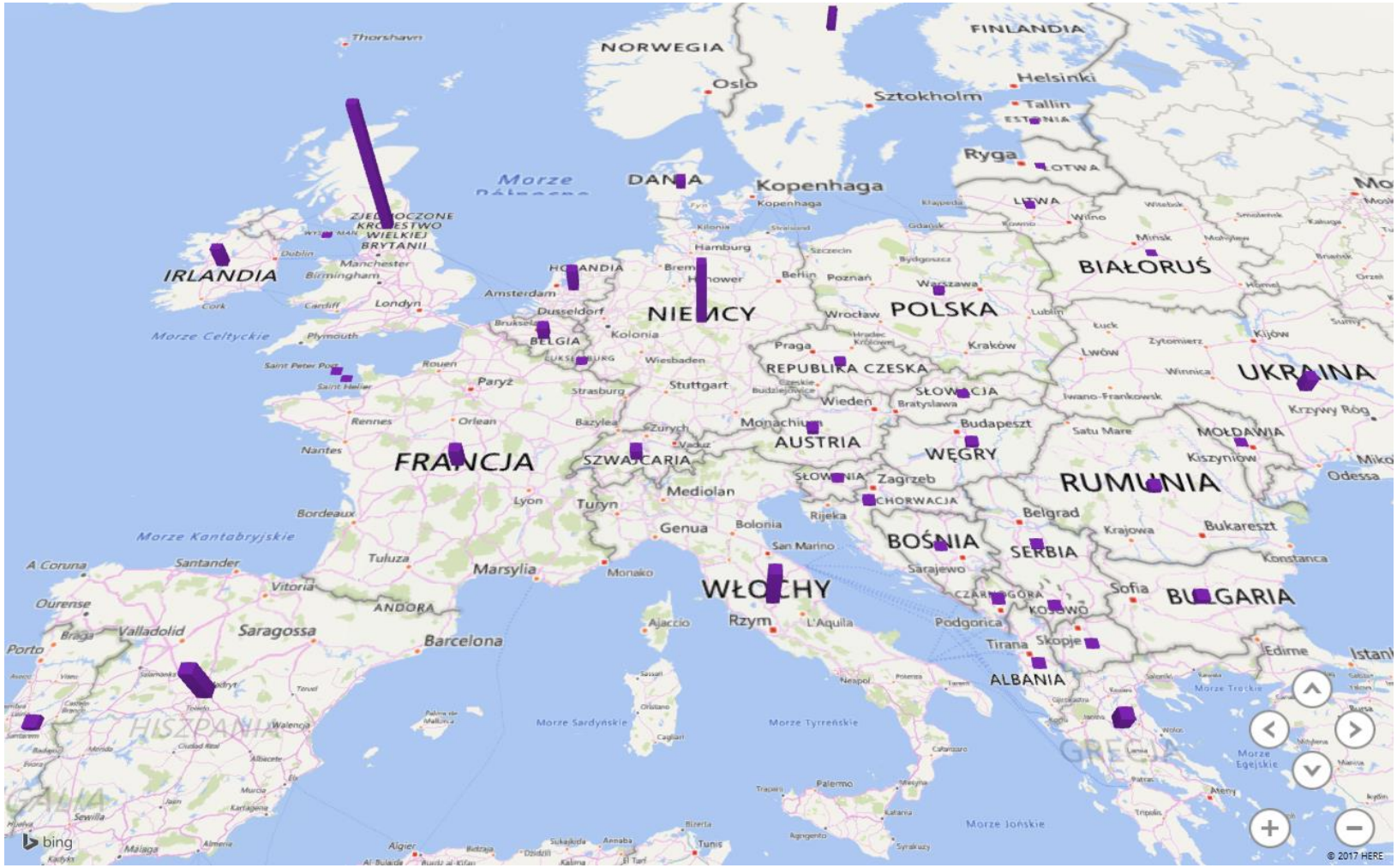
Table . Estimated number of trips made by Polish residents in air traffic in 3rd quarter of 2017.

Country	Travels	Country	Travels	Country	Travels
Albania	1 145	Germany	286 844	Nigeria	98
Algeria	401	Ghana	59	Norway	140 022
Angola	70	Gibraltar	212	Oman	132
Argentina	170	Greece	61 791	Pakistan	4
Armenia	2 831	Guadeloupe	12	Panama	53
Aruba	10	Guernsey	36	Portugal	21 516
Austria	19 417	Hong Kong	460	Puerto Rico	3
Azerbaijan	64	Hungary	18 592	Qatar	9 261
Bahamas	6	Iceland	9 501	Reunion	104
Bahrain	27	India	514	Romania	20 927
Barbados	0	Indonesia	28	Russia	28 941
Belarus	4 592	Iran	209	Rwanda	12
Belgium	54 980	Iraq	77	Saudi Arabia	314
Benin	34	Ireland	87 245	Senegal	56
Bermuda	1	Isle of Man	54	Serbia	3 621
Bosnia and Herzegovina	317	Israel	29 897	Seychelles	51
Brazil	501	Italy	166 059	Sierra Leone	16
Bulgaria	24 792	Jamaica	16	Singapore	433
Burkina Faso	36	Japan	5 250	Saint Maarten	28
Cameroon	32	Jersey	136	Slovakia	5 014
Canada	34 309	Jordan	199	Slovenia	3 948
Cape Verde	32	Kazakhstan	3 070	South Africa	358
Caribbean Netherlands	10	Kenya	97	South Korea	6 322
Central African Republic	15	Kosovo	197	Spain	149 996
Chad	22	Kuwait	128	Suriname	24
Chile	63	Latvia	6 971	Sweden	110 905
China	10 417	Lebanon	214	Switzerland	46 196

Colombia	61	Lithuania	8 321	Taiwan	132
Congo (Brazzaville)	30	Luxembourg	4 475	Tajikistan	3
Congo (Kinshasa)	13	Macau	3	Tanzania	20
Cote d'Ivoire	64	Macedonia	304	Thailand	442
Croatia	12 145	Madagascar	23	Togo	16
Cuba	802	Malaysia	47	Trinidad and Tobago	6
Curacao	32	Maldives	2	Tunisia	1 506
Cyprus	12 476	Mali	25	Turkey	39 665
Czech Republic	15 008	Malta	23 481	Turkmenistan	4
Denmark	45 639	Martinique	16	U.S. Virgin Islands	1
Djibouti	16	Mauritania	12	Ukraine	61 732
Dominican Republic	1 904	Mauritius	97	United Arab Emirates	26 972
Ecuador	14	Mayotte	30	United Kingdom	578 365
Egypt	5 926	Mexico	1 199	United States	55 637
Equatorial Guinea	1	Moldova	2 953	Uzbekistan	16
Estonia	4 382	Mongolia	15	Venezuela	18
Ethiopia	147	Montenegro	3 061	Vietnam	98
Finland	26 790	Morocco	1 850		
France	72 400	Namibia	58		
Gabon	26	Netherlands	96 573		
Georgia	8 556	Niger	25	TOTAL	2 494 152

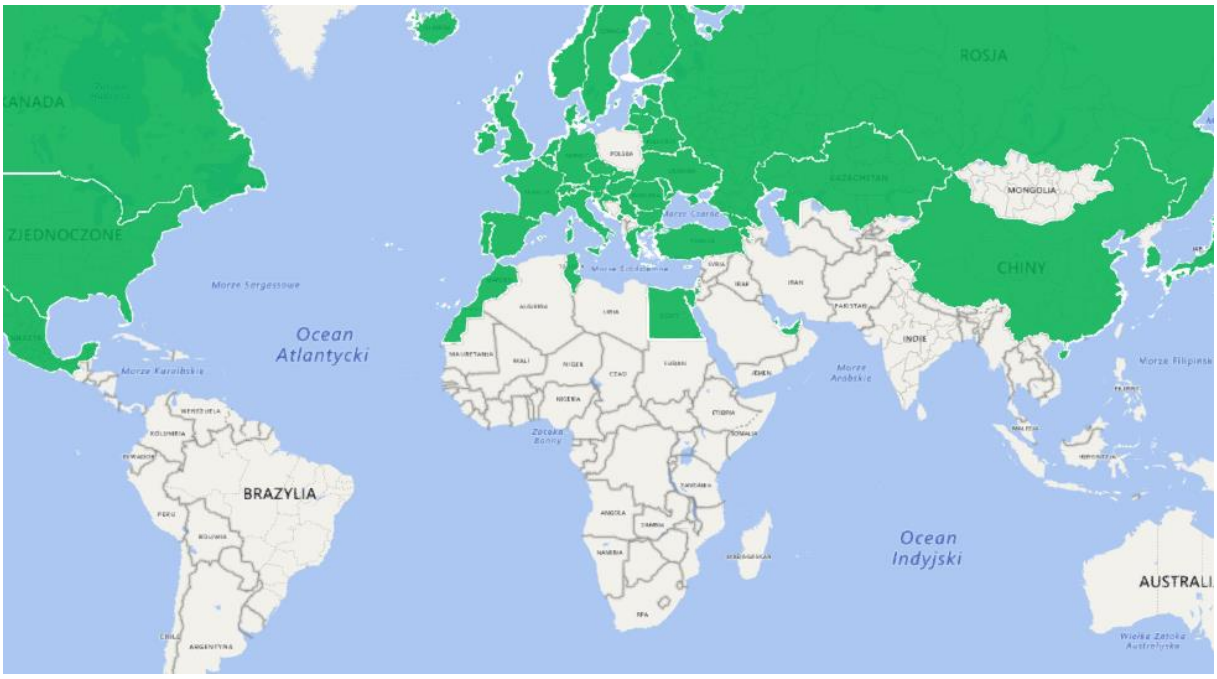
Source: Own elaboration

Graph. Distribution of trips made by Polish residents in air traffic in 3rd quarter of 2017 in Europe

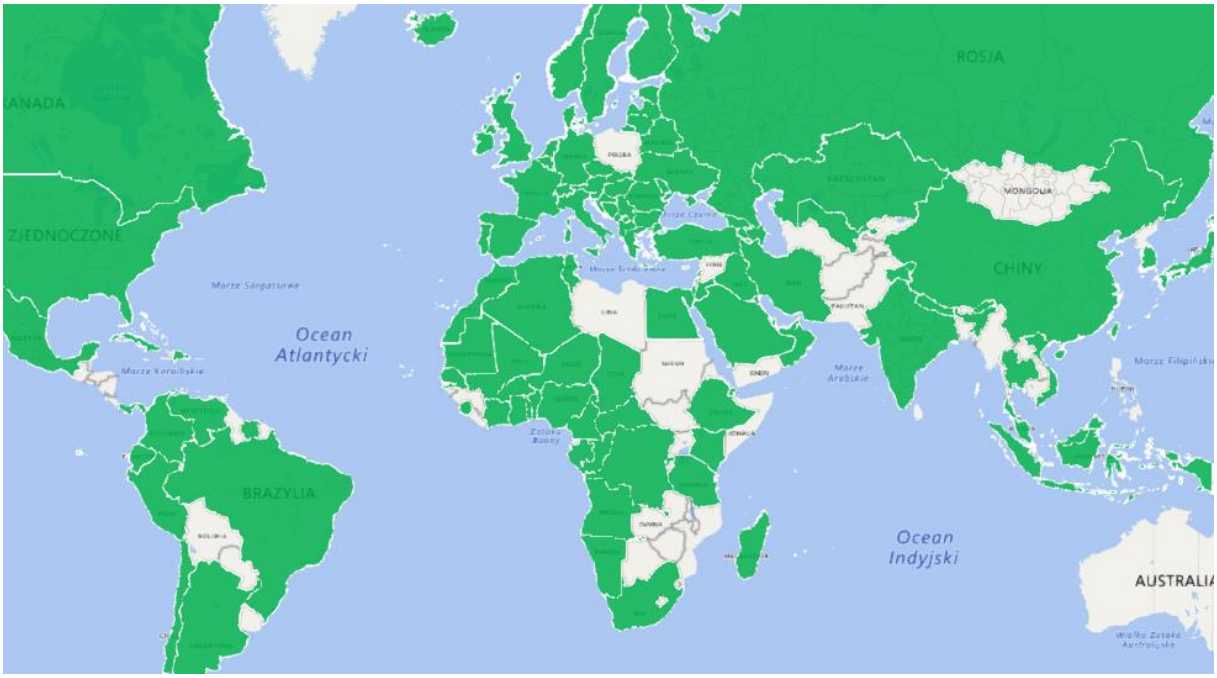


To present an advantage of using Big Data we attach two maps of coverage – with green colour they are marked countries with non-zero estimate. The first one with directs flight only and the second one after Big Data application.

Graph. Countries with non-zero estimate from direct flights.



Graph. Countries with non-zero estimate from direct flights and through hubs.



Source: Own elaboration

Border Traffic Compilation

Air traffic and land border traffic have completely different data sources and methods. Nevertheless, the outcomes can be integrated. With administrative data from Border Guards on land border traffic on external EU's border, data from Maritime Offices on maritime border traffic and Big Data it can be compiled a table on border traffic. Some statistics pertaining to share of non-residents in total traffic were used from sample survey. Big Data from traffic loops was used to estimate land border traffic on internal EU's border while Big Data from web scraping of flight schedules was used to estimate air traffic with respect to the first visited country.

Table. Border traffic with respect to country and dimension (land, sea and air) in 2015

	Number of crossings		
	Total	Polish residents	Non-residents
Total	217 175 084	88 121 949	129 053 135
land border	187 972 038	70 960 914	117 011 124
external EU's border	35 031 906	6 371 720	28 660 186
Russian-Polish border	6 098 860	3 369 138	2 729 722
Belarusian-Polish border	7 817 371	863 830	6 953 541
Ukrainian-Polish border	21 115 675	2 138 752	18 976 923
internal EU's border	152 940 132	64 589 194	88 350 938
Lithuanian-Polish border	4 946 268	1 528 605	3 417 663
Slovakian-Polish border	15 271 001	6 864 510	8 406 492
Czech-Polish border	50 870 006	24 917 817	25 952 189
German-Polish border	81 852 856	31 278 263	50 574 593
maritime border	1 851 524	1 088 040	763 485
air border	27 351 522	16 072 995	11 278 527
external EU's border	1 037 256	611 388	425 867
Russian-Polish border	315 118	185 740	129 378
Belarusian-Polish border	49 997	29 469	20 527
Ukrainian-Polish border	672 141	396 179	275 962

internal EU's border	3 947 888	2 326 998	1 620 890
Lithuanian-Polish border	90 604	53 405	37 199
Slovakian-Polish border	54 588	32 176	22 412
Czech-Polish border	179 694	105 917	73 777
German-Polish border	3 623 002	2 135 501	1 487 501
other air borders	22 366 379	13 134 608	9 231 770

Source: Own elaboration

Train trips

Passenger train transportation slowly recovers after decade of decreasing number of passenger. In the period 2014-2016 total number of passenger increased by 8.7% reaching 292.6 mln. of passengers in 2016. In 2017 the expected number of passenger will be over 300 mln.

Train companies improved a lot their websites providing new services and information. For instance, there are available:

- Timetables
- Location of coaches
- Occupancy indicator
- Forecasts of frequency

Timetables covers around 180 domestic and 11 foreign connections. Each route has full information on arrival and departure time for every station. Search of train connections for some of them provides information on occupancy in the form of icon as follows:

Source: https://www.intercity.pl/pl/site/dla-pasazera/informacje/frekwencja.html

For instance, the first one means that most of the seats are available, the second one means that more than half is already booked. Moreover, passengers may find information on forecasts of frequency in trains as a percentage of seats occupied:

- Over 80% seats occupied
- 50%-80% seats occupied
- Under 50% seats occupied

What is more, data on location of coaches is available. Composition of trains are presented in example below

Source: https://www.intercity.pl/pl/site/dla-pasazera/informacje/pociagi-i-stacje/rozklady-jazdy/

With that it is possible to estimate number of seats in each class and with certain facilities (sleeping wagon, wagon with berths etc.).

Statistics on total number of passenger are available with delay of 2-3 months. With extrapolation of trend it can be used as a benchmark for further estimations. Since in a timetable we observe connections but not passenger behaviours we need to combine it with sample survey where passenger behaviours can be extracted to some extent. It is similar approach to combining Bigdata on air traffic with sample survey.

Nevertheless, domestic train transportations seem to much more complicated than air traffic.

4.3. Main findings

Border traffic and tourism seem to be perfect statistical domains to be combined with Big Data. Our preliminary research show huge potential and usefulness of Big Data.

To conclude:

- Data on road traffic intensity may decrease a burden of interviewers because data on border crossings can be estimated,
- Data obtained has a better spatial granularity than from sample survey,
- A great care need to be taken in choosing sensors with respect to their location, elsewhere data may by heavily biased,
- Missing data imputation needs completely non-standard approach. Cross-entropy-based approach seems to be relevant,
- Big Data on air traffic may increase countries coverage for which statistics are produced,
- Estimates are much more stable and distributions are smoother than in sample survey,
- Nevertheless, aircraft schedules cannot be used solely or even with administrative data. Some crucial statistics still need to be obtained from sample survey.
- Big data produces much more Europeanised trips distribution rather than country-specific. Therefore, distributions derived from BigData approach need to be integrated with distributions derived from sample survey which requires relevant methods,
- Train traffic data has a potential to be used to enhance domestic tourism statistics but it much more complex task than other presented cases

Big Data can be implemented in Integrated Survey of Tourism conducted by Statistical Office in Rzeszów. Road sensor data from Lithuania has been already implemented in border traffic estimation since 2018.

Next steps can be undertaken in Big Data usage research. From our experience of this study future direction of research are:

- Full integration of Big Data and sample survey. Estimation of impact of Big Data on all variables in sample survey – not only number of trips,
- Develop methods for combining distribution received from Big Data and distribution received from sample survey.

5. Agriculture

5.1. Introduction and overview

The purpose of the work was to use Big Data in the mechanism for recognizing selected crops in Poland (Table 1) on the basis of satellite images, administrative data and field surveys (in-situ). The remote sensing data came from Sentinel satellites operating under the European Copernicus program. A time series of satellite data representing the development of crops in 2017 were used, which were processed and then classified. The reference data were databases on crops that were collected during In-situ studies and information from administrative systems, including: handling the direct payments to farmers - Land Parcel Identification System (LPIS).

Table 8 Selected crops

N°	Crop
1	Winter wheat
2	Spring wheat
3	Rye
4	Winter barley
5	Spring barley
6	Oat
7	Winter triticale
8	Spring triticale
9	Mixed of winter cereals
10	Mixed of spring cereals
11	Maize
12	Winter rape
13	Spring rape

Radar data processing was elaborated on the base of Sentinel-1 A & B data acquired in Stripmap Mode with the VH/VV polarization. From the beginning of the year 2017 more than 2200 scenes were downloaded for the whole country (3.75 TB). Data were downloaded in a fully automatic way. Surface of Poland is covered by 6 paths of Sentinel-1 satellites (Figure 1). Five paths were used in the study because there were not enough training points for the 6th path. Each path is covered by 2-4 image frames. Paths overlap in 30-40% of their surfaces. For all paths, at least 32 dates of different images were acquired. For further analysis 18 dates

were selected (Annex 1) according to the class separability analysis done on data from previous years.

Figure 1 Paths of Sentinel-2 satellites analysed for the surface of Poland

Sentinel-2 images were taken for two purposes. The first objective was to support image segmentation, and the second to better distinguish winter from spring crops. A total of about 1.2 TB of data was downloaded (over 1160 images). Based on the series of optical data for the period: autumn 2016 - spring 2017, the NDVI vegetation index (Normalized Difference Vegetation Index) was calculated. Then a multi-layer mosaic was created to obtain a continuous NDVI layer for the whole country, complementing the cloudy areas on the last spring photo with the NDVI values from previous spring, winter or autumn photos.

The following administrative administrative sources were used as reference data:

- cadastral parcels vector data from Land Parcel Identification System (over 34 million records, 13GB of data) - used for in-situ plots selection and also used for satellite data segmentation,

- agricultural plots vector data from Land Parcel Identification System (over 33 million records, 23GB of data) - used for in situ plots selection and used for satellite data segmentation,
- General Geographic Database (BDOO)

and supporting data from:

- SRTM (Shuttle Radar Topography Mission) - areas with a slope of more than 3 °,
- vector borders of the SENTINEL-1 satellite imaging paths.

As reference data, an in-situ crop database was used, which was prepared on the basis of field information gathered by experts in agricultural areas throughout Poland from June to August 2017. Over 5000 plots (Figure 2) have been identified that meet the criteria for the field survey dedicated for this study.

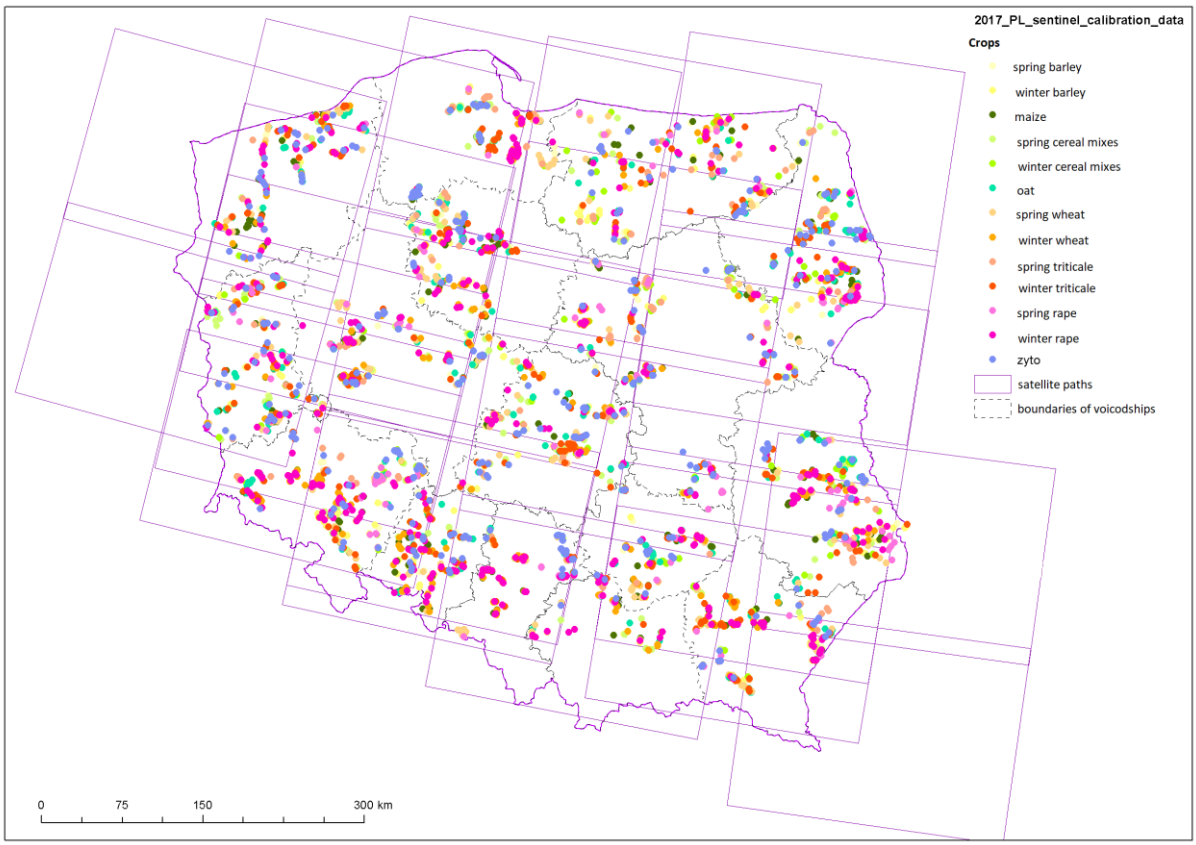


Figure 2. Distribution of samples of particular crops by the voivodships and Sentinel-2 flight paths

System learning and object classification was performed on in-situ data:

- sample data geodatabase – 5084 records (fig.3),
- image geodatabase – 21747 geotagged images, 32GB of data

Crop	Number of samples
spring barley	481
winter barley	340
corn	509
spring cereal mixes	448
winter cereal mixes	207
oat	426
spring wheat	432
winter wheat	559
spring triticale	218
winter triticale	448
spring rape	134
winter rape	476
rye	406
Total	**5084**

Figure 3. Number of fields for each crop.

The same pilot but with smaller number of crop types was conducted by Statistics Ireland. For better accuracy, only four different crop types were used in the training dataset. Other crops were undetermined.

5.2. Intra-domain analysis

The object based image classification also uses the mosaic of NDVI vegetation index calculated on the basis of Sentinel-2 multi-sensed optical images. The first stage of object based image classification was segmentation based on the division of the image content into the so-called - objects, that are then classified. Segmentation has been done based on multitemporal SAR images supported by Sentinel 2 optical images. Segments were limited to the borders of agriculture lands specified in the LPIS database. In the first phase, LPIS vector was used for the exclusion of the areas that are outside of the agriculture lands (e.g. forests, water, built-up areas). First level of the segmentation was done based on the LPIS vector. Within the agricultural plots, segmentation on radar layers was made using the multiresolution segmentation method. It allowed the creation of segments that distribute non-homogeneous crops well within one field. Then, the objects underwent re-segmentation

using the channels from optical images S2: R (red), NIR (near infrared) and SWIR (short infrared), to refine the segments (Fig.4).

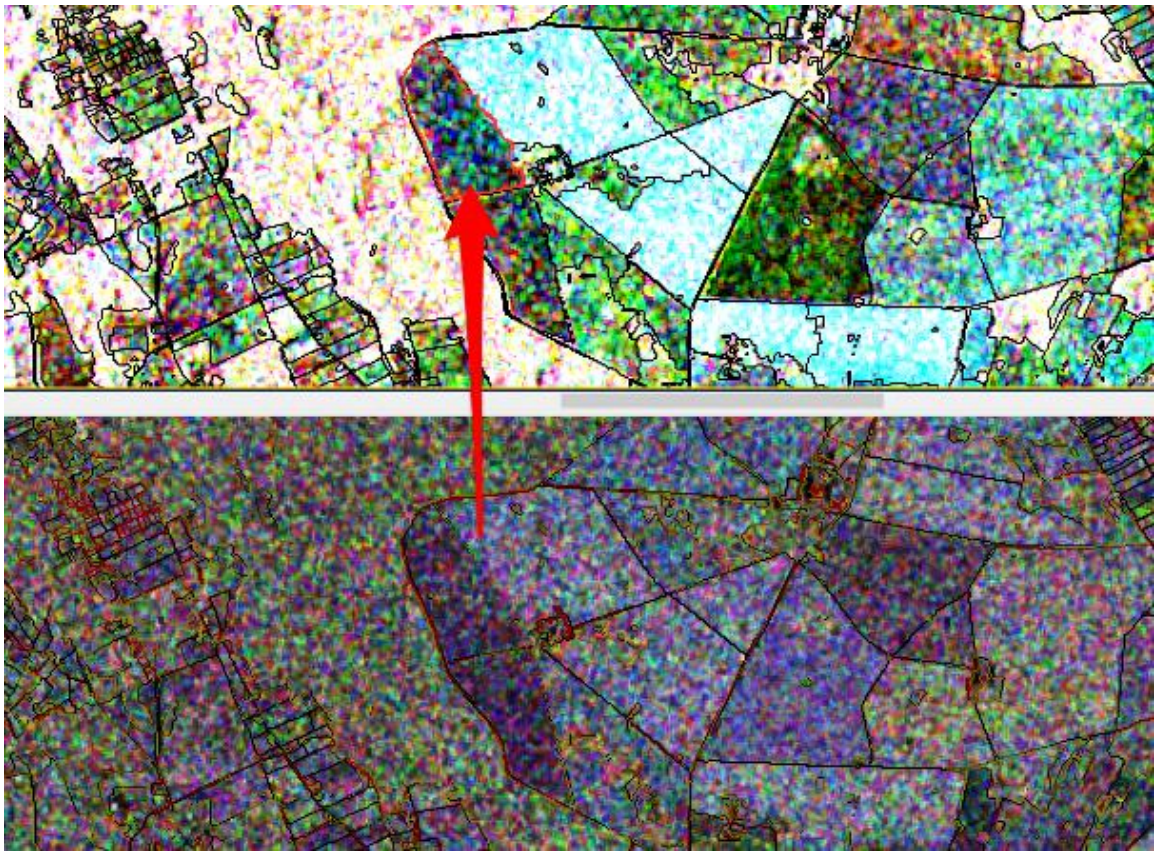
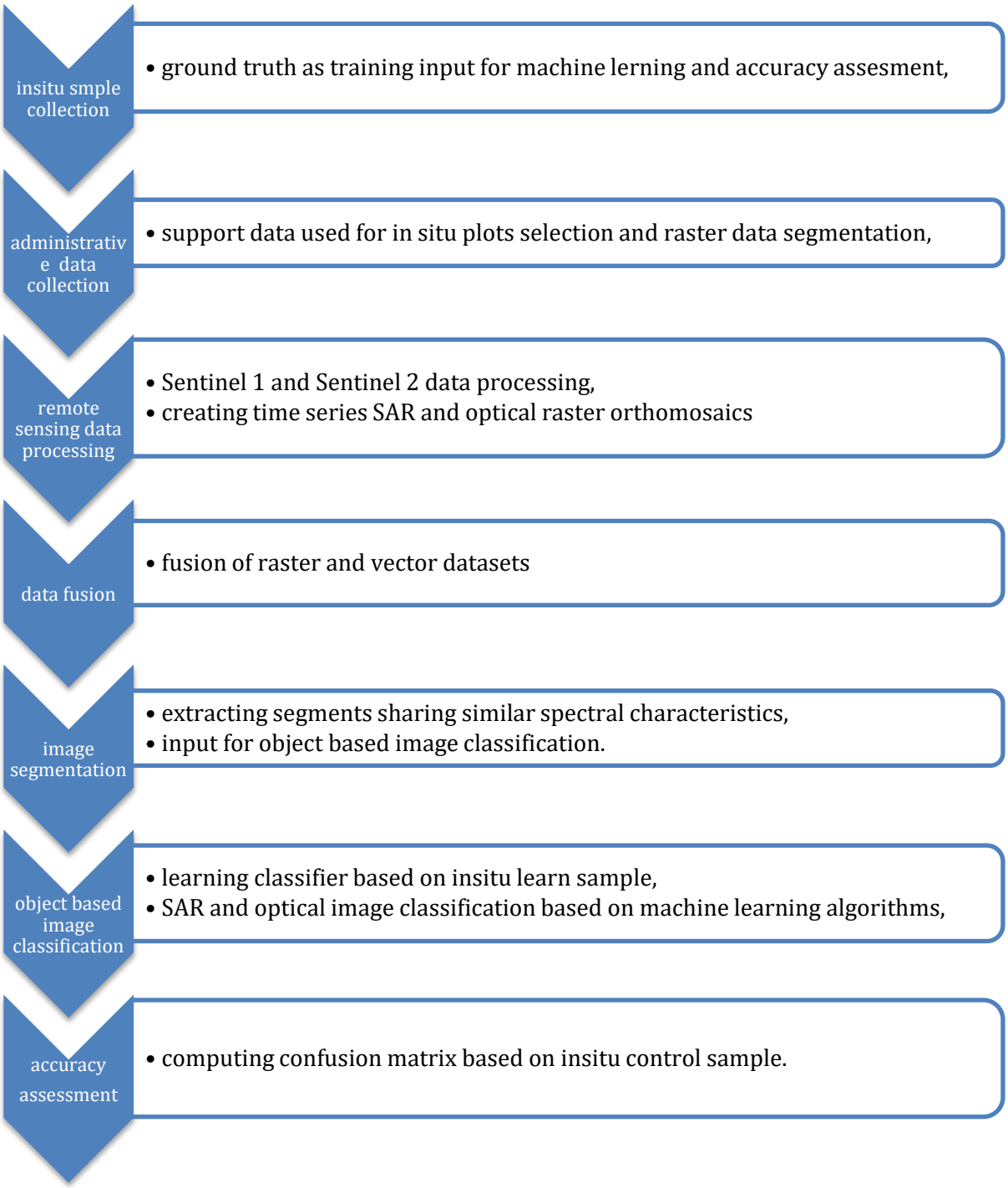


Figure 4. Results of segmentation, multitemporal Sentinel1 SAR images. Up: segmentation results, down: LPIS vector layer

Then, on the extracted segments, the object based image classification was made using the machine learning algorithms (Support Vector Machine) and the areas of individual crops in poviats (NTS 4) and voivodships (NTS 2) were calculated (Annex 2). The process of data analysis and processing is presented in the diagram below:

Data processing workflow:



5.3. Main findings

During the project Integration of administrative data, field surveys and satellite images were used in the scope of:

1) source data - preparation of a training set - combining georeferenced vector data with satellite and in-situ data to prepare machine learning algorithms;

2) result data - the ability to publish results from satellite images and combine them with data from administrative data, e.g. in areas where there is no information in administrative sources.

This is not a registration number or other descriptive attribute of the parcel but georeference (spatial reference) of these data is the main attribute (data linkage) for combining administrative data with satellite, while combining spatial data from different sources is called as data fusion. Unfortunately, there may be even several different crops on one record plot, which prevents their full reference mapping with remote sensing data, hence the use of the image segmentation method. The biggest problem encountered during the analysis of optical data from the Sentinel 2 satellite was cloudiness (Fig.5 and Fig.6). In the context of the optical imaging of the terrain surface, it is important to use every observation opportunity. In principle, this should translate into an increase in the number of cloudless observations.

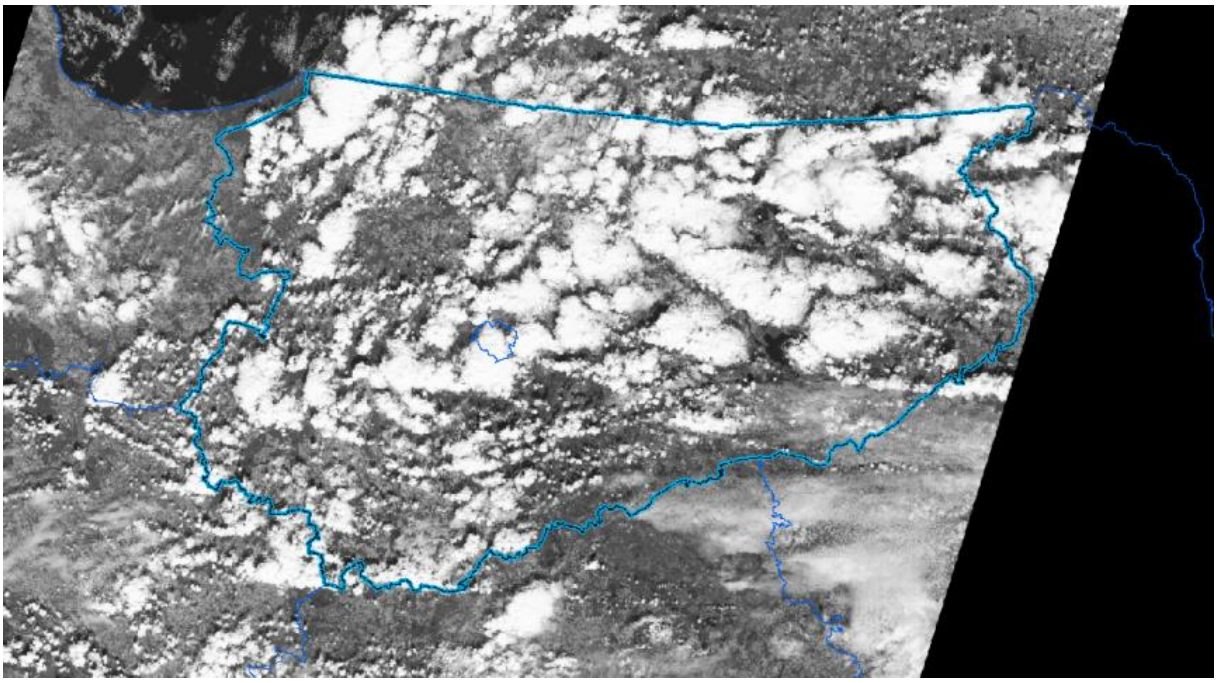


Fig. 5. Coverage of the warmińsko-mazurskie voivodship by the Sentinel-2A observations including cloud cover (white areas).

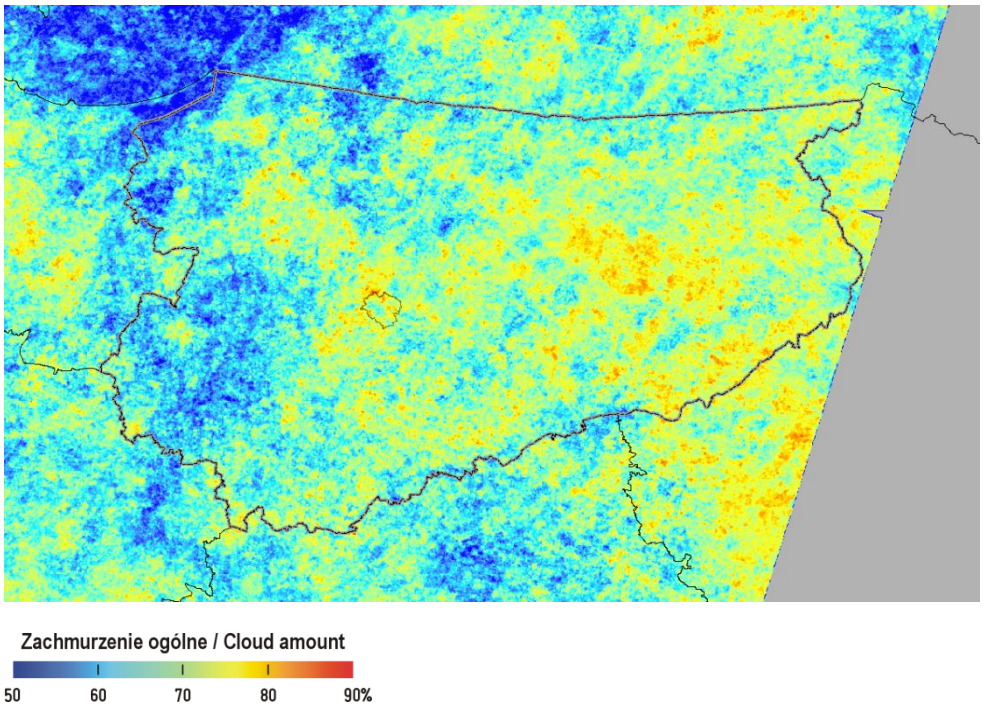


Fig. 6. Average annual cloud cover, determined on the basis of Sentinel-2A observations

The obtained results allow to formulate the following conclusions:

<u>The resulting data:</u>

- **The general distinction between classes is good**, but in some cases the separation of morphologically similar plants is a problem (spring wheat vs spring triticale).
- The use of the multitemporal NDVI mosaic for the autumn 2016 - spring 2017 period allowed a better distinction of winter and spring cereals.
- The machine learning **accuracy** varies from **80% to 97**%. A detailed analysis of the causes of differences in the machine's learning accuracy is required, most of which

result from the differences in field sizes in individual regions of Poland, which affects the accuracy of machine learning of the system.

- **Segmentation carried out in both radar and optical images is better than that carried out only in radar images.**
- When starting work related to the recognition of crops using satellite data, each type of crop should be represented by about 50 points in the field survey and have a relational mapping in the administrative data.
- Due to the specificity of radar data processing, the field of field surveys should be defined in accordance with the distribution of picture frames, not administrative borders.
- The addition of previous autumn images to the series of data allows for a better distinction between spring and winter crops and improves the overall accuracy of the classification.
- Due to the spatial resolution of the Sentinel-1 data, it is not possible to correctly classify arable fields smaller than 60x60 m at this time.

Data combining:

- Connecting data at the source stage - administrative and in-situ data as well as result - recognition of crops, requires time-consuming and advanced system work, which is associated with a significant load of IT systems with Big-Data.
- **The use of data fusion satellite and administrative data allows to increase the accuracy of published results.**
- The combination of various remote sensing and administration data allows in individual cases to eliminate errors in administrative data and improve their quality.
- The fusion of satellite and administrative data creates the possibility of preparing and developing the estimated and resulting data much earlier than in comparison with the results of statistical surveys carried out in the field of cropping areas.

6. Inter-domain analysis

Agritourism

Rural areas in Poland have significant touristic potential in the form of rich natural resources, culture and infrastructure. In addition, agritouristical farms continue to expand their offer addressed to the most demanding guests - better accommodation, additional attractions such as horse riding, fishing, tennis courts, summer pools, bonfires or regional events.

In the third quarter of 2017 Polish people made 18.7 million of domestic trips. Cottage house or agritouristical lodging was chosen in 2.7% of trips what gives around half million of trips. Sample size does not allow to produce results on NUTS 2 level. For each NUTS 2 region coefficient of variation exceeds 20% (on average it is 47%). One of region even did not appear in a survey. Therefore, we cannot state anything on expenses, trips and nights spent etc. on NUTS 2 level without relevant support of external sources.

We suppose that landscape and natural resources are crucial in choosing the best place for agritouristical activity. Thus, information on landscape structure may be used. Number of agritouristical lodgings is a derivate of landscape and natural resources. Distribution of agritouristical lodgings seems to be also valuable.

The aim of this study is to produce estimates of expenses, trips and nights spent on NUTS 2 level with better precision than from sample survey solely. Following data sources are planned to be used:

- Farm Structure Survey (NUTS 2 level) conducted by Central Statistical Office in Poland
- Integrated Survey of Tourism conducted by Statistical Office in Rzeszów (NUTS 2 level)
- Land and Buildings Register from Central Office of Geodesy and Cartography (NUTS 5 level)
- Lowland / highland structure (NUTS 3 level)
- Data from web scraping of webpages on agritouristical objects

Integrated Survey of Tourism is the source of information on expenses, trips and nights spent. Farm Structure Survey is a sample survey which covers data on location of homesteads, legal status (a natural person, a legal person etc.), presence of economic activity other than agriculture or agritourism. Land and Buildings Register provides data on landscape structure. Number of agritouristical lodgings is obtained from web scraping of relevant websites.

Since direct estimations on NUTS 2 level are not reliable and there are no additional variables in a tourism survey that totals or means are known in general population there is very limited number of methods that can be applied. At the beginning we used synthetic regression estimator. It turned out that OLS regression produces inadmissible outcomes (negative values). We needed a tool that is less sensitive to unreliable dependent variable and may shrink result towards mean. One of these is lasso regression. Details about that methods are described in attachment. Regression results are presented below.

Table. Lasso-based estimates on agritourism in Poland in the third quarter of 2017.

NUTS 2 region	Nights spent	Visitors	Total expenditures	Expenditures on accommodation
dolnośląskie	157299	49154	38063176	14679858
kujawsko-pomorskie	136604	36714	29430374	10747222
lubelskie	70444	22043	16848246	5726209
lubuskie	128360	21880	16659689	5419688
łódzkie	59530	11610	11132149	3784183
małopolskie	287019	93947	78408199	31192865
mazowieckie	8560	13869	2554044	174132
opolskie	98382	13508	16270084	5796491
podkarpackie	236122	73300	59136618	23019659
podlaskie	42345	4429	3984058	443955
pomorskie	187210	51759	38114586	14071677
śląskie	103083	20885	17365269	6587026
świętokrzyskie	100661	16825	18306975	6588852
warmińsko-mazurskie	190729	60776	42748426	15414115
wielkopolskie	91135	32685	18947858	6505026
zachodniopomorskie	226147	68740	47743768	17433312

Source: Own elaboration

7. Legal, quality and technical aspects

7.1. Legal aspects

According to the data sources used, there are no restrictions that will not allow official statistics to use them. For most of the use cases official API's were used to retrieve the data. For road sensors data we established cooperation with main government authorities responsible for the data to deliver data for official statistics purposes. Therefore, in the pilots conducted there are no legal issues that will not allow us continuing the development of the pilots in the future.

7.2. Quality

In WP7 three topics were investigated: sentiments revealed in the social media; agriculture; tourism/border crossing. Sentiment analysis is based on the posts from social media. People that are active on social media may differ from the total population and their number and activity can change over time. So the representativeness is an issue when using this data source. This makes a challenge to compare the results with the current population. Also extracting background characteristics of the users and accurately determining the sentiment of the text not easy tasks. Depending on the social media channel, in some countries Twitter is not so popular and used only by selected group of people. The accuracy is measured with machine learning algorithms and varies between 60 and 90%, depending on the training dataset and country (pilot was conducted by Poland, Portugal and UK).

In the field of agriculture, WP7 combined satellite images with administrative data and survey data to train the machine learning algorithms to recognize different crop types. Sources can be linked accurately, findings can be verified by manual inspection of land lots and the first results are very promising. The accuracy is measured by the number of fields with crop types identified correctly and varies from 75% to 85% depending on the crop type and machine learning algorithm used (KNN and SVM are the most accurate). The pilot was conducted by Poland and Ireland, using different approaches.

In the Tourism/border crossing domain WP7 developed a methodology to measure intensity of border traffic by the number of vehicles/airplanes and passengers. Data about border traffic was obtained from government authorities in Poland and neighbouring Schengen member countries. Moreover, data on air traffic and train trips was scrapped from web portals to see the whole perspective of border

traffic movement. This enable comparison of the traffic from both sides of the border. At this moment the results of traffic movement by road sensors can be presented for the border between Poland and Czech Republic, Lithuania, Germany and Slovakia. The estimates show number of vehicles, people/vehicle and expected vehicles. More detailed information is for air traffic – the number of countries is significantly higher because of the specific information available on the aircraft type and source and destination airport. The accuracy is an issue with the road traffic as there are some gaps in the data and some data must be estimated – it leads to the possibility of providing the data under or overestimated.

7.3. IT issues

Although we have conducted several different use cases, there are some specific areas in which we can find a common point. For example, in population domain we had used widely Python language in most of the use cases conducted. Below you can find tools and methods used.

Table 9. Tools used

Use case	PL	PT	UK
Population/Twitter Life Satisfaction	Python with Pandas, Numpy, Sklearn MS Excel with CSV files	Python with Pandas, Numpy, Sklearn MS Excel	Python with Pandas, Numpy, Sklearn MS Excel
Population/Facebook Guardians people's moods			Python (for NLP the following lexicons were used: VADER, Syuzhet, Afinn, Bing, NRC) MongoDB
Tourism/Tourism Accommodation establishments	Python, CSV files		
Tourism/Border Crossing	Python R GAMS MS Excel with CSV files		

Tourism-Agriculture/Agricultural places	Python MS Excel, CSV files		
Agriculture/Crop types identification	CSV, PDF MTSar ArcGIS		

The following Big Data sources have been taken into processing, according to the table below.

Table 10. Data sources used in pilots

Source description	Structured Unstructured	Source	Processing	Metadata Integration
Big Data	Unstructured	Satellite	Machine Learning	The training fields selection criteria Data segmentation Data classification Data aggregation Results assessment (quality of output)
Big Data	Unstructured	Web	Web scraping (flight movement)	Aggregation
Big Data	Unstructured	Social Media	Machine Learning	Automatic classification of the source
Big Data	Structured	Road sensors	Entropy Econometrics	Clustering, Estimating
Big Data	Unstructured	Web	Web scraping, Machine Learning	Automatic classification of the source

8. Conclusions

The pilots conducted in the ESSNet Big Data WP7 resulted in many benefits to NSI's. It includes:

- we can have an accurate data from satellite images with the high accuracy,

- we know the rules of using API's and processing the data from social media, such as Facebook and Twitter,
- we can get the current data from road government authorities from road sensors for different countries and we know how to process them when there is no data,
- we can get data by web scraping alternative data sources, such as air and train traffic,
- we have better skills on machine learning regarding sentiment analysis and satellite data processing.

There are still few challenges:

- our results show that it is not possible to support population use cases (e.g., counting population) because the data sources available at this moment are not representative,
- web scraping tourism accommodation establishments makes it possible to identify all the objects, but the classification used by websites for tourism is not unified (e.g., hotels vs. pensions),
- we have to work constantly on improving algorithms as the data sources are not stable and its structure and format may change (e.g., resolution of satellite images).

To summarize:

- WP7 prepared and tested 6 intra-domain pilots in three different domains (3 Population, 2 Tourism, 1 Agriculture with 2 pilots implemented by different approaches, first by Statistics Poland and the second by CSO Ireland).
- The most tested pilot and most promising in Population domain is Life Satisfaction – it uses machine learning algorithm to produce the results of life satisfaction according to classification from EU-SILC survey (happy, neutral, calm, upset, depressed, discouraged), based on Twitter data.
- Other two pilots in Population domain are related to the selected health status of population (Morbidity areas use case on depression (Google Trends) by ONS UK) and to Peoples opinion/interestingness by topics based on websites by ONS UK by Facebook.
- Alternative use cases on Population include the possibility of estimating the density of population by BTS location (Base Transceiver Stations used by mobile phone operators) and daily and night population by Google Traffic, but they were not implemented at international level in the first wave of pilots.

- In Tourism there are two different pilots: Tourism accommodation establishments and Internal EU Border Crossing, including data sources by Air Traffic – Flight Movement web scraping and Traffic Loops data.
- Agriculture domain has one pilot prepared by two different methodological approach : first was prepared by Statistics Poland and the second was prepared by CSO Ireland.
- For data combining we have two different approaches – intra-domain data combining (each domain) and inter-domain data combining (agriculture-tourism).
- The results of machine learning algorithms for agriculture domain have an average accuracy of 80% (the percentage of positively identified crop types), while in pilots for population domain also varies between ~55% and 80% (the percentage of positively identified life satisfaction class – e.g., upset, calm, happy, depressed, …), depending on the algorithm and training dataset.
- The results of the pilots conducted shows that the great potential is in the Agriculture domain – to identify crop types. The case is ready to use with the open data that can be accessed on the Internet.

Appendix 1. Tourism Domain

1. Road traffic model

The method that was applied to achieve the main goal of the use case is an adaptation of crucial elements of entropy econometrics. It consists of two stages: data combining and data imputation. In the first stage we focus on time series point where at least two sources of data is available.

Latent true value of traffic Y is observed indirectly by data sources e.g. S_1 and S_2. Each source of data has a random error component with normal distribution assumed. In the most basic form model conditions are described as following

$$Y = S_1 + \epsilon, \ (1)$$

$$Y = S_2 + \eta \ (2)$$

where $\epsilon \sim \mathrm{N}(0, \Omega_\epsilon), \ \ \eta \sim \mathrm{N}(0, \Omega_\eta)$.

If any exogenous variables X are available then one more source of information can be added in a form of $X\beta$ as a regression term

$$Y = X\beta + \xi, \ (3)$$

where $\beta \sim \mathrm{N}(\beta_0, \Omega_\beta)$, $\xi \sim \mathrm{N}(0, \Omega_\xi)$.

Model parameters ϵ, η, ξ and β can be estimated with Conditional Generalized Least Square method with variance-covariance matrix of the form

$$\Omega = \begin{bmatrix} \Omega_\epsilon & 0 & 0 & 0 \\ 0 & \Omega_\eta & 0 & 0 \\ 0 & 0 & \Omega_\xi & 0 \\ 0 & 0 & 0 & \Omega_\beta \end{bmatrix}$$

More generalized approach employs idea of cross-entropy, concept from information theory. The cross entropy is a measure of similarity of distribution q to distribution p. It is determined by the Shannon formula

$$H(p, q) = -\sum_{i=1}^{n} p_i \cdot \log q_i$$

It achieves the minimum value when $p = q$. The cross-entropy is based on the concept of distribution. We shall rewrite the model as follows

$$Y = S_1 + \epsilon = S_1 + \textstyle\sum_i p_i \epsilon_i,$$

$$Y = S_2 + \eta = S_2 + \textstyle\sum_j q_j \eta_j,$$

$$Y = X\beta + \xi = X \textstyle\sum_k b_k \beta_k + \sum_l r_l \xi_l,$$

where ϵ_i is a support of probability measure p_i etc. and

$$\textstyle\sum_i p_i = \sum_j q_j = \sum_k b_k = \sum_l r_l = 1, \ \ p_i, q_j, b_k, r_l > 0.$$

We are looking for new distributions p', q', r', b' by minimizing

$$H(p', p) + H(q', q) + H(r', r) + H(b', b)\,.$$

with respect to conditions (1)-(3).

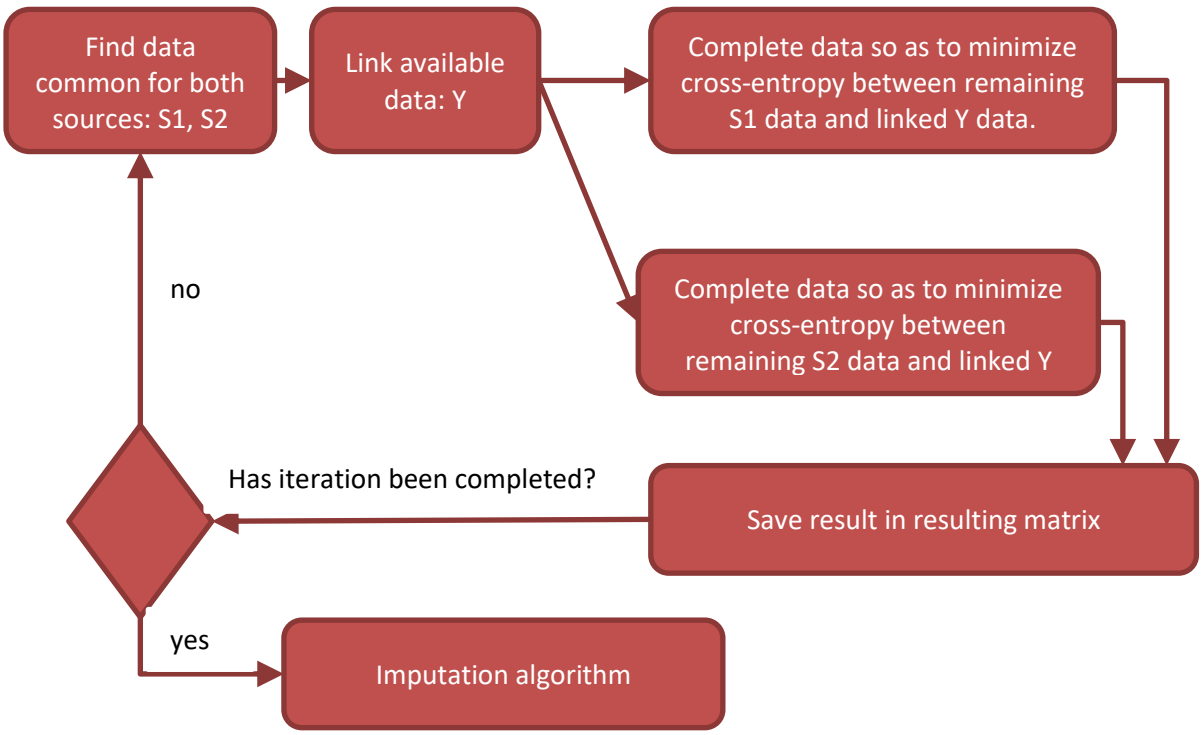
Model provides reference points for each time series with at least two sources of data for point of time which are the result of combining data sources. Reference points are the final estimates of road traffic.

The next step enables to estimate data when reference points are available and there are remaining points in the sources. Let $(x_0, \dots, x_n)$ be time series from the source and $(y_1, \dots, y_n)$ be time series of reference points. We are looking for missing value z. It can be found by minimizing

$$H(z) = -\sum_{i=1}^{n}\left(\frac{x_i}{\sum_{i=0}^{n} x_i} \cdot \log\frac{y_i}{z + \sum_{i=1}^{n} y_i}\right) - \frac{x_0}{\sum_{i=0}^{n} x_i} \cdot \log\frac{z}{z + \sum_{i=1}^{n} y_i}$$

In fact we complete the data by minimizing cross-entropy between source and reference source obtaining new value in reference source. The same procedure for the second source.

Whole procedure is presented in diagram below.



After saving the results for all of the time series we start to deal with points with no data. At first, we look for missing point of data. Remaining points in time series is called ‘base row’, remaining points in given year is called ‘base column’. We check the similarity between

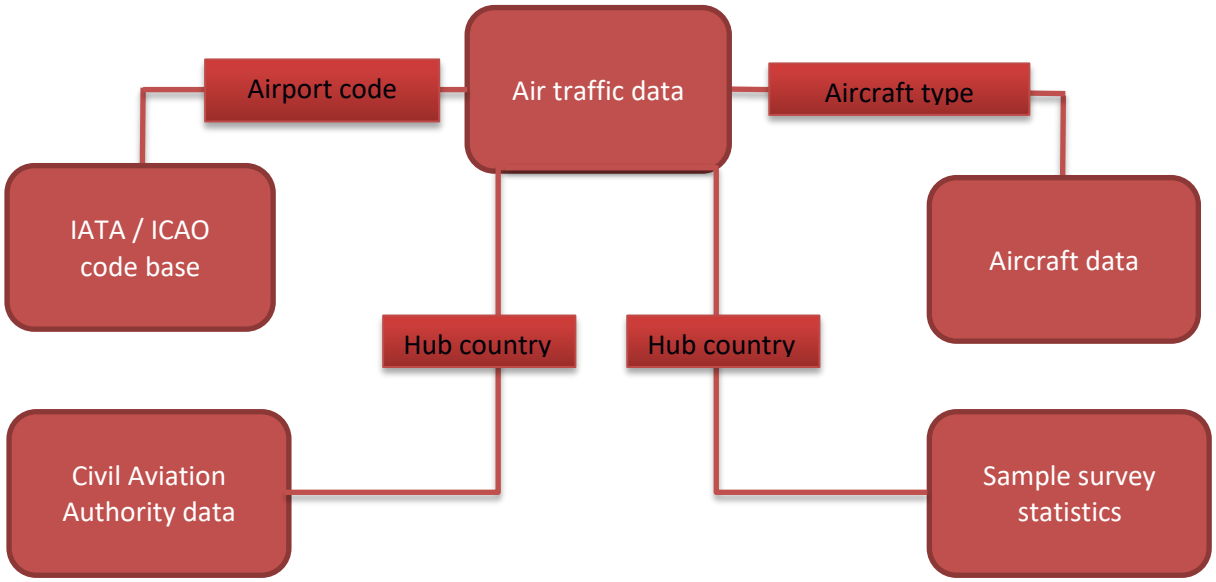
base row and every row in data base as well as for base column. We pick the best row or column basing on some simple validation criteria and cross-entropy measure. The best one (row or column) is defined as an optimal vector. Then we impute data gap by minimizing cross-entropy between base vector and optimal vector.

2. Air traffic model

The main part of modelling is data linkage. Three sources: air traffic data, aircraft data and IATA/ICAO code base were linked by keys: IATA/ICAO code and aircraft type. After data each flight had assigned capacity, origin and destination country. Origin/destination table was calculated with pivot table by summing up capacities of each flight on relevant route. For each hub country all of the estimates were scaled to benchmark to figures from Civil Aviation Authority. Then we applied two statistics: percent of Polish resident in total international air traffic and percent of Polish resident using hub airports to retrieve number of Polish resident in air traffic and distribute them into two categories: fly further or not from each hub country. That gave us a number of passenger from each hub country flying further and it allowed us to use distribution of passenger from each hub country.

Diagram below presents data sources and key that was used to link each source.

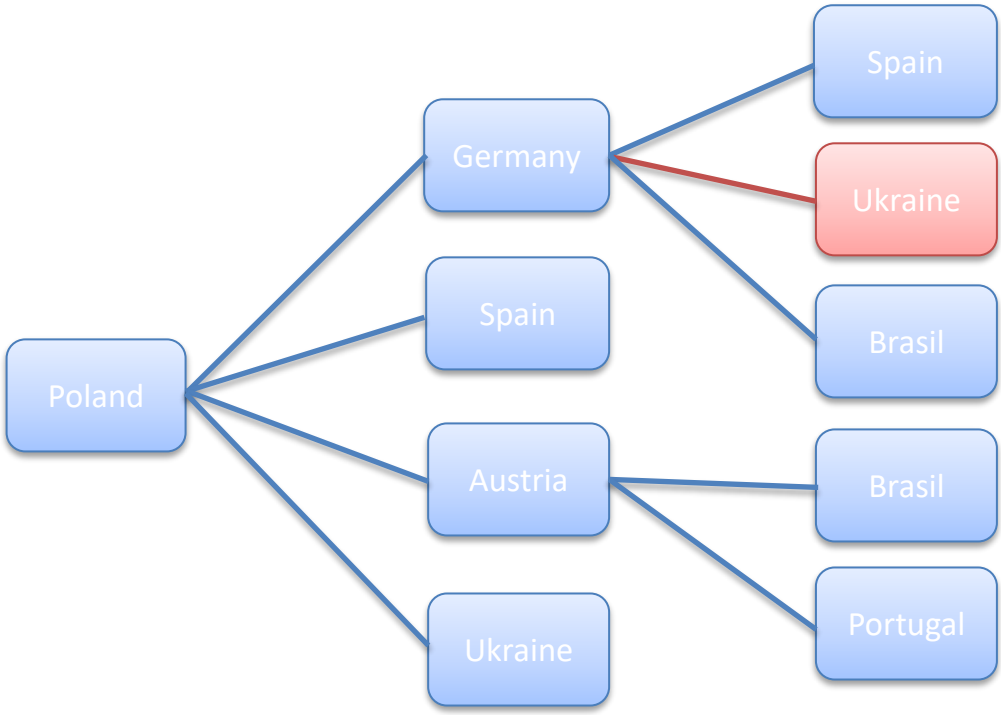
Data linkage in air traffic.



Since we have data on full distribution of flights we needed to exclude those routes that would not be used by Polish passengers for sure or with high probability. For instance, we excluded all routes back to Poland e.g. Poland-Germany-Poland (Poland is not a final destination). Also we excluded not very probable routes such as Poland-Germany-Ukraine

since there are lot of direct flight from Poland to Ukraine and Poland-Germany-Ukraine route is not very time or cost efficient. Schema below presents idea of route selection. With orange colour excluded route is marked.

Route selection in air traffic.



Annex 1 The terms of classified Sentinel-1 images

Path 1	Path 2	Path 3	Path 4	Path 5
20170417	20170418	20170419	20170420	20170415
20170423	20170424	20170425	20170426	20170427
20170429	20170430	20170501	20170502	20170515
20170505	20170506	20170507	20170508	20170521
20170511	20170512	20170513	20170514	20170527
20170517	20170518	20170519	20170520	20170602
20170523	20170524	20170525	20170526	20170608
20170604	20170530	20170531	20170601	20170620
20170610	20170605	20170606	20170607	20170626
20170616	20170611	20170612	20170613	20170702
20170622	20170617	20170618	20170619	20170708
20170628	20170623	20170624	20170625	20170714
20170704	20170629	20170630	20170701	20170807
20170710	20170705	20170706	20170707	20170831
20170716	20170711	20170712	20170719	20170912
20170809	20170717	20170811	20170812	20170912
20170902	20170810	20170904	20170805	
20170914	20170915	20170916	20170917	

Annex 2. Crop structure in voivodships

	Winter wheat	Maize	Mixed of spring cereals	Winter rape	Spring barley	oat	Winter triticale	Winter barley	Mixed of winter cereals	Rye	Spring wheat	Spring triticale	Spring rape
	2	3	4	5	6	7	8	9	10	11	12	13	14
Dolnośląskie	173253	145881	47073	154136	59687	41720	47594	39244	14211	42753	49929	11192	18243
Kujawsko-pomorskie	106582	221238	99912	136147	62746	37729	67247	56993	19979	90421	36283	19340	18053
Lubelskie	80226	134882	54770	104419	61947	54242	127243	16081	53798	87439	79212	68082	51356
Lubuskie	37762	53809	44258	52875	25441	21838	44464	19008	9486	32579	21665	12868	11471
Łódzkie	82661	127058	117794	49828	81216	86792	68448	36679	34115	87062	43037	19569	21436
Małopolskie	73606	55655	18170	30125	39699	14592	26300	21019	19546	14241	31585	7411	13414
Mazowieckie	128534	285548	93481	79358	106266	94890	115691	98596	61298	123378	72642	62513	28074
Opolskie	79370	60109	19870	66634	24564	11434	28393	20131	7224	23074	19036	5672	9106
Podkarpackie	43720	48561	12391	27083	21252	25955	37957	27630	13067	33414	43715	28357	40337
Podlaskie	57053	148827	72773	22294	44103	64786	59429	23586	31177	78835	51738	29961	12161
Pomorskie	94457	80850	62211	108989	49716	56912	65305	21061	16989	55091	49370	14028	9681
Śląskie	63784	42242	27681	31258	19671	21060	27327	9276	9928	18534	28313	7033	7325
Świętokrzyskie	71479	55883	28618	21855	44362	29265	23556	19767	10970	15941	39932	6231	9594
Warmińsko-mazurskie	160519	121593	62298	114711	48849	31884	95813	34177	29862	78641	74727	36332	17037
Wielkopolskie	130171	323677	149718	152008	120466	88570	135491	94627	27284	161912	50577	28547	29495
Zachodniopomorskie	90595	86818	66665	136496	56379	45216	64045	26356	11254	82772	78270	22716	26544

3. Web scraping tourism websites

Under Big Data ESSNet WorkPackage 7, the Portuguese Team from Statistics Portugal, decided to extract reviews from tourism related web portals and compare the results with official statistics from tourism surveys. The main benefits and added value for official statistics would be to provide timely insights of the phenomena that are difficult to measure through traditional surveys.

For this purpose the team applied webscrapping, datamining and machine learning techniques in order to compare data for tourism flows, visitor's country of residence, gender and age group.

Source selection

Given the growing demand for Lisbon as a tourist destination, we decided to choose a tourist attraction in this city that met two conditions:

a) Important in number of visitors and diversity of nationalities;

b) Well documented on the web (through reviews, ratings, etc.).

We considered several tourism web portals before committing to one in particular:

- https://www.tripadvisor.com
- https://www.yelp.com/local_yelp
- https://pt.foursquare.com
- https://maps.google.com/localguides
- https://www.trip.skyscanner.com/pt
- https://www.flipkey.com
- https://www.expedia.com
- https://www.airbnb.pt
- https://www.booking.com
- https://www.priceline.com

These webportals have some differences as well some similarities. They all have a staggering amount of users and reviews and also have some form of rating. Some are more business related and more social netting focused, such as Yelp, others require an account and login just to access the information (such as Google Local Guides). TripAdvisor is probably the most popular webportal in Portugal in its category although at this time we don't have hard data to support this. All things considered we opted for TripAdvisor. TripAdvisor Travelers' Choice Award (that nominates the best destinations and attractions based on the opinions and user ratings) awarded to Oceanário de Lisboa (Best Aquarium in the World 2015 and 2017) have led us to choose this tourist attraction.

Oceanário is considered the tourist attraction number 1 in Lisbon, receiving more than 1 million visitors a year, with a daily record of 8,909 (August 17, 2017).

Since 1998, the opening year, 20 million people of 180 nationalities have visited it.

The Oceanário page on Tripadvisor has about 30,000 reviews written in 26 languages.

Based on these facts, it seemed obvious to choose this source for our exploratory study. However, since the Oceanário is also visited by many national tourists, we decided to extract information from

two other attractions that "were even more touristic" and that we could compare with the Oceanário – Tram28 and Mosteiro dos Jerónimos; for these two attractions, the data extraction was decided and carried out in the data exploration phase of the Oceanário reason why the information extracted is not so extensive.

According to a survey from Studies and Statistics Office of Turismo de Lisboa for 2016[3], monuments and museums are the main reason (85.1%) foreign tourists visit Lisbon. Data from TripAdvisor reveals that Mosteiro dos Jerónimos is the most reviewed monument in that webportal.

Information available on TripAdvisor pages

We can group the information on TripAdvisor pages in:

- Information related to the review – review date, title, text, rating, helpful votes, review language, device used to produce the review,
- Information related to the user who created the review - username, user geographical origin, total number of written reviews, contributor level, TripAdvisor member year, age, gender, number of visited cities and number of uploaded photos,
- Information related to the performance of the user, as a member of TripAdvisor (badges, reviews, etc.) that we do not find so pertinent to this study.

The username, also referred as “screen name” is not, unfortunately a unique identifier for the author of the review. The email of the user could provide that unique key identifier, but that information is not publicly available. Therefore is possible to have identical “screen names” for different users which in turn may have an undetermined impact especially if we intend to combine reviews for multiple tourist attractions for the same city. Ultimately we would need to combine all the information available for users such as reviews, number of cities visited, user ranking geographical origin, etc… to produce a unique identifier. This way we could deal more effectively with duplicates.

Reviews can be submitted with a time-lapse up to a year and have to be approved before becoming available online. The users can write more than one review for each item (monument, tourist attraction, restaurant…) but with a mandatory 3 month interval. This is relevant if considering the use of this data to produce timely statistics, for ex. Tourist flows for the previous weekend.

As the analysis was totally exploratory we decided to extract all the information described in a) and b) available on the site for the selected attractions. Note that:

- There are variables for which information is not always available (examples: user geographical origin, language identification, gender, age, etc.),

For Tram28 and Mosteiro dos Jerónimos, due to timing constrains, it was only possible to extract data from reviews written in English. Some profile data such as contributor level, TripAdvisor member year, age, gender, number of visited cities and uploaded photos were also not extracted.

[3] https://www.visitlisboa.com/sites/default/files/2017-05/Inqu%C3%A9rito%20Motivacional%202016%20-%20Cidade%20de%20Lisboa_0.pdf

Web data extraction

To extract data from the Oceanário page on TripAdvisor we used web scraping with scripts purposely developed in the Python language.

The data volume to be extracted and the coexistence in the site of static and dynamic contents (which require the use of different extraction techniques) led to the following extraction strategy:

Phase 1 - Extraction of the list of review addresses (url)

To interact with language selection and page navigation handling, we used Selenium with Firefox; Selenium is a portable software-testing framework for web applications, which creates automated instances of browsers, simulating the user's interaction with the website through the browser. Is a good tool when it is necessary to interact with web pages (click buttons, impute values in search fields, etc.) and therefore very good to deal with Javascript.

Phase 2 – Extraction of static content for each url review

In this phase we used Scrapy, a framework for extracting data from websites designed to make multiple requests at the same time enabling faster extractions.

10 variables were extracted - review date, title, text, rating, helpful votes, review language, device used to produce the review, username, user geographical origin, total number of written reviews.

Note that:

- review text - only texts written in English were extracted in this phase; texts written in other languages are dynamically generated and therefore were extracted in the 3rd phase,
- user geographical origin is an open string and for that reason totally unstructured (can have anything, such as locality, district, region, country or all this mixed) and is not available for all reviews,
- language review variable is also not available for all reviews.

Phase 3 - Extraction of dynamic content for each url review

In this phase, we handled dynamic extraction of content for each url review, which returned:

- Review text for other language than English,
- Content that requires interaction with the site through the “user popup”.

5 variables were extracted - contributor level, TripAdvisor member year, string variable with age and gender included, number of visited cities and number of uploaded photos.

At this stage we used again Selenium and Firefox; the need for interaction with the site (after each url request), the amount of data to be extracted and compliance with the rules of netiquette (especially time request) made this phase slow and in need of constant monitoring (we had be alert for any possible internet connection or any site blockage). This 3rd phase, as said before, was not executed for Tram28 and Mosteiro dos Jerónimos due to time constraints.

In addition to the extraction code, the scripts also included some text cleaning code with regex (whitespace, new lines, handling numbers, etc.).

Preprocessing and classification tasks

The preprocessing was done also in Python, in a Jupyter Notebook environment.

We can divide the processing and classification tasks done in two levels:

1. Cleaning and normalize tasks (reducing noise and extract useful data from strings) – convert to lower case; remove accents; remove punctuation; remove whitespaces and tabs; remove duplicates; extract variables from profile strings (age, gender); handling numbers and dates, etc.,

2. Classification tasks

1. User geographical origin

As we said previously user geographical origin is an open string totally unstructured (it can have anything, such as locality, district, region, country or all this mixed available… without a stipulated and identifiable order). Facing this limitation we decided to use the Geopy python package to geolocalize this string and extract the country. This package provides several classes that can be used for this purpose (Google, Bing, OpenStreetMap, etc.); however we wanted to privilege open data and then we used the OpenStreetMap class Nominatim. The process was slow because we worked online; but as it is open data in future uses we can work offline.

A second approach was then attempted using sequential approaches and subject matter knowledge. Official statistics and data produced from relevant stakeholders such as Lisbon tourism Association provided the latter.

The sequential approaches to classify the country of origin where:

- Data cleaning
- Data Extraction (REGEX)
- Exact matching of the resulting list of cities with a complete database for:
    - Country name
    - Country code,
    - Population,
    - Continent name,
    - EU membership
- Rule based classification
- Classification with other extracted variables (language)
- Data cleaning included the removal of noise (reference to age group, gender), extraction of country and / or city when available.

In some cases this data cleaning was enough to identify the country, for other cases the result was the city or city and the state / province.

Exact matching was then performed by city to impute country. Duplicate city names either for the same country or different countries had to be addressed. For instance, there are 7 cities named "Lisbon", 6 of them are located in the United States of America. Another example, there are a total of 11 cities named "Paris" located in France, Canada, Panama and United States. In some cases the duplicates are even more frequent when considering cities with less than 1000 inhabitants. For this task we only used data from cities with more than 1000 inhabitants. If duplicates persisted, variables such as population and EU membership were used to select the appropriate result. The results were then human supervised.

Rule based classification consisted mainly in resolving specific frequent strings resulting from the previous steps. For example, states from particular relevant countries such as Massachusetts and Texas for the United States and Belo Horizonte and São Paulo for Brazil were classified with this rule based approach.

Next we tested the variable "language" to identify the country. Languages such as Portuguese, English, French or Spanish were excluded from this approach due to the number of countries that have one of those languages as an official language. English is obviously widely used regardless of the country of origin of the user. For languages such as "it" and "pl" this process is pretty much straightforward (even though there are some known exceptions). For others like "nl" or "ru" we found that could be imputed to at least 2 countries each based on the distribution of our webscrapped data. Therefore it would require some form of weighting for those imputations which was not performed at this stage. Official statistics provided valuable information when assessing the use of this approach since we only considered the most frequent languages and the most relevant countries of origin.

This approach – the use of contextual variables to impute the country of origin - requires careful consideration because it can perform very well for some countries but may fail short on others.

For this second approach to identify user geographical origin we used R Packages (tidyverse, janitor and stringi) and all the tasks were performed offline.

2. Language review

This variable is available for about 98% of Oceanário reviews. Despite the 2% not available we found that the wealth of a dataset with almost 30,000 reviews written in 26 languages should be used; then, we decided to train a model of supervised machine learning that would enable us to classify the missing 2%.

For the supervised machine learning test we used Python and the SciKit-Learn library in a Jupyter Notebook environment.

For feature extraction we used a vectorizer that splits review strings into sequence of 1 to 6 consecutive characters (according SciKit-Learn site suggestion we used characters instead of word tokens to better capture the language specificities – "fingerprints"); these groups of characters were our ngrams.

Then we used the linear classifier Perceptron with the previous vectorizer; we fit the classifier on a training set of 80% and predict the outcome on a testing dataset of 20%.

The test produced a median accuracy of around 99% and 0,15% std, a precision = 99%, a recall = 99% and a f1- score=99%.

Finaly we used the classifier to classify the 561 reviews without lang variable.

As validation we used Langdetect python package (library ported from Google's language-detection) on the same 561 reviews and compare the results; we had an agreement rate of 99,6% (only two reviews had different languages attributed by the two methods).

Following these promising results, we tried to classify the same 561 reviews without the lang variable with the R Package "cld3" (Google's Compact Language Detector 3), a neural network model for language identification. Although still experimental, the successor of cld2 performed very well with an agreement rate of 100% when comparing with the supervised machine learning algorithm results.

Data Analysis

Comparing tourism official statistics with data scrapped from tourism related web portals like TripAdvisor is challenging task and one that comes with some risks. While the former is produced for this specific purpose (tourism statistics) the later only reflects the TripAdvisor user's reviews on certain tourist attractions. These two sources don't use the same concepts; hence plain common sense advises against drawing definite conclusions from this comparison. Nevertheless this analysis falls in the category of exploratory analysis.

There was an effort to reduce the gap between these two data sources and to some extent this was possible by confining data to specific parameters:

- Time-frame: the year 2016
- Geographical localization : Metropolitan Area of Lisbon (NUTS II)
- Population: Non residents (Portuguese tourists were excluded)

Guests in Tourism Accommodation in Metropolitan Area of Lisbon by Place of Residence

As reference for official data, we selected Guests in tourism accommodation located at Metropolitan Area of Lisbon (NUT II) by place of residence (non-residents only) for 2016. Tourism accommodation activity is presented for overall establishments' hotels and similar, rural tourism and lodging tourism and finally Local accommodation (10 or more beds).

Data from TripAdvisor, namely, data from reviews on Oceanário, Jerónimos Monastery and Tram 28 was previously presented.

For comparison purposes we used percentage values.

Next table presents data from the two sources (Official Statistics and TripAdvisor) in percentages. Data is sorted by the ten most represented places of residence according to official statistics. Aggregates were created in order to reduce the amount of information displayed.

Place of Residence	Official Stat AML	Oceánario	Jerónimos	Tram28
France	13,1	15,8	16,4	21,3

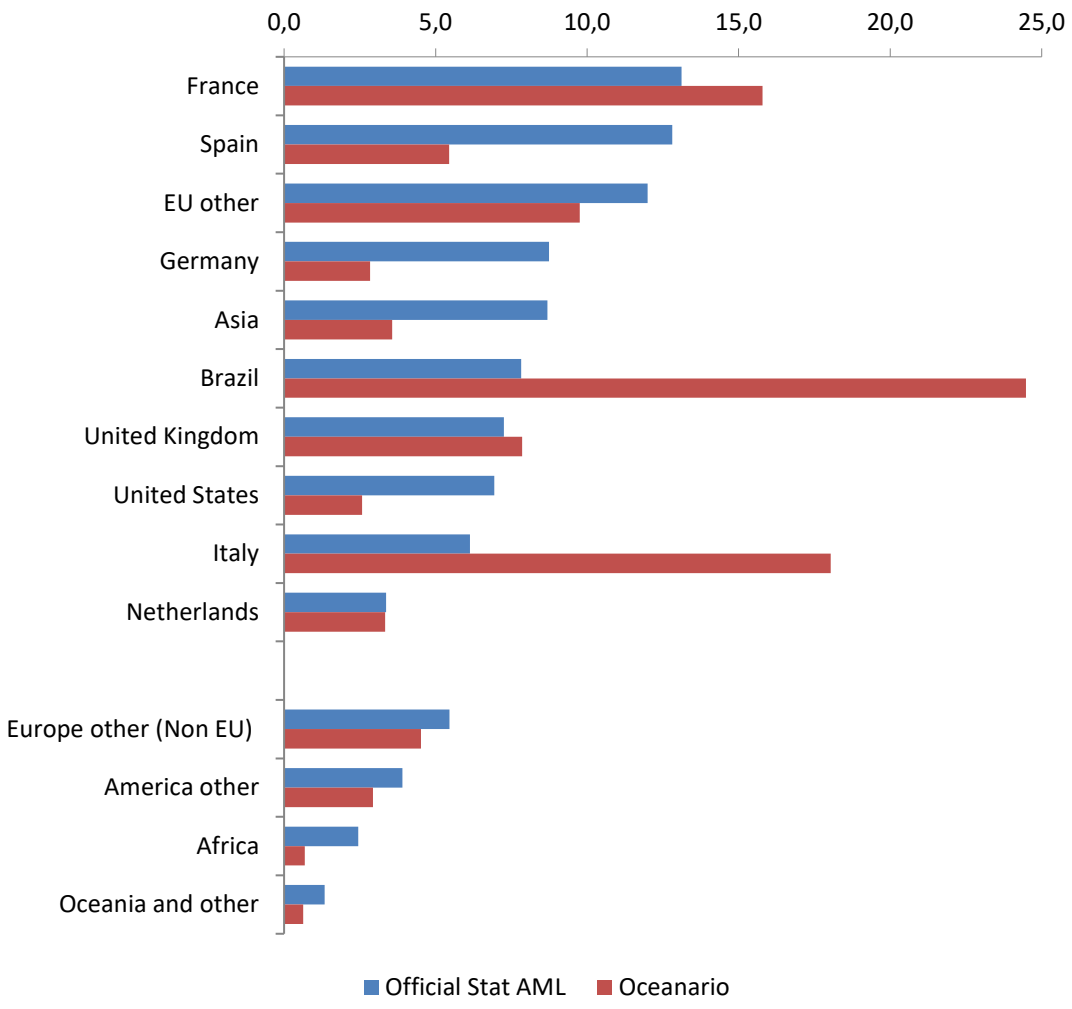
Spain	12,8	5,4	8,7	6,9
EU other	12,0	9,8	13,8	13,1
Germany	8,7	2,8	2,2	4,0
Asia	8,7	3,6	6,8	5,1
Brazil	7,8	24,5	20,5	6,6
United Kingdom	7,3	7,9	6,9	9,3
United States	6,9	2,6	4,5	3,1
Italy	6,1	18,0	18,6	22,2
Netherlands	3,4	3,3	2,8	5,3
Europe other (Non EU)	5,5	4,5	3,0	3,8
America other	3,9	2,9	5,6	4,1
Africa	2,4	0,7	0,5	0,6
Oceania and other	1,3	0,6	1,2	1,5
%	100,0	100,0	100,0	100,0
N	4419833	5600	5344	4418

When comparing official data and TripAdvisor user reviews' by country of residence we can observe relevant differences for Brazil and Italy as they are over represented.

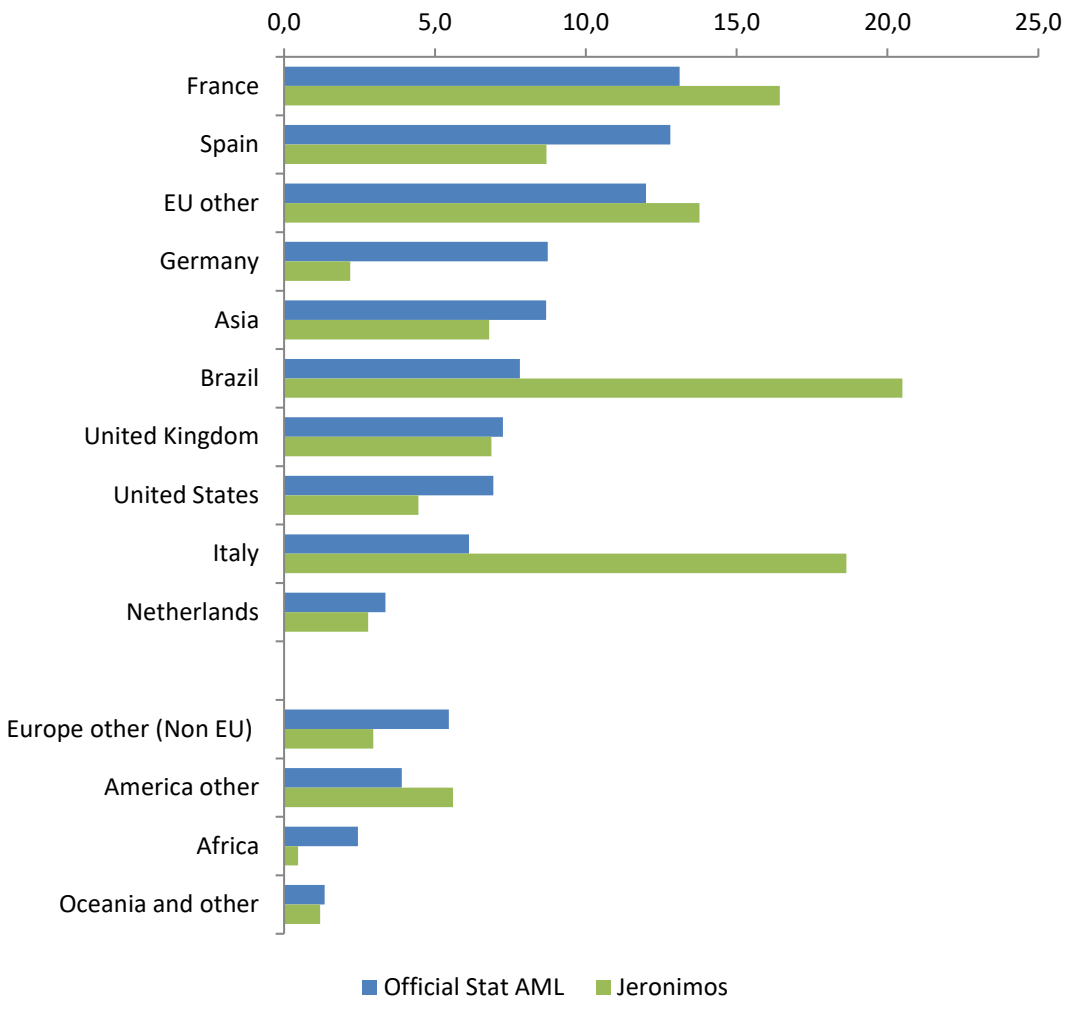
Reviews from Brazilian users account for 24,48%. This value may be explained on account of the large Brazilian community that lives in Portugal and particularly in the Metropolitan Area of Lisbon, but probably identifies themselves as "from Brazil" in their TripAdvisor's profile. At this moment we do not have an explanatory hypothesis for relevant percentage of Italian tourists.

Tourists from Spain, Germany, United States and "Asia other" appear underrepresented in TripAdvisor's data.

Country of residence: Official statistics and Oceanário (TripAdvisor)

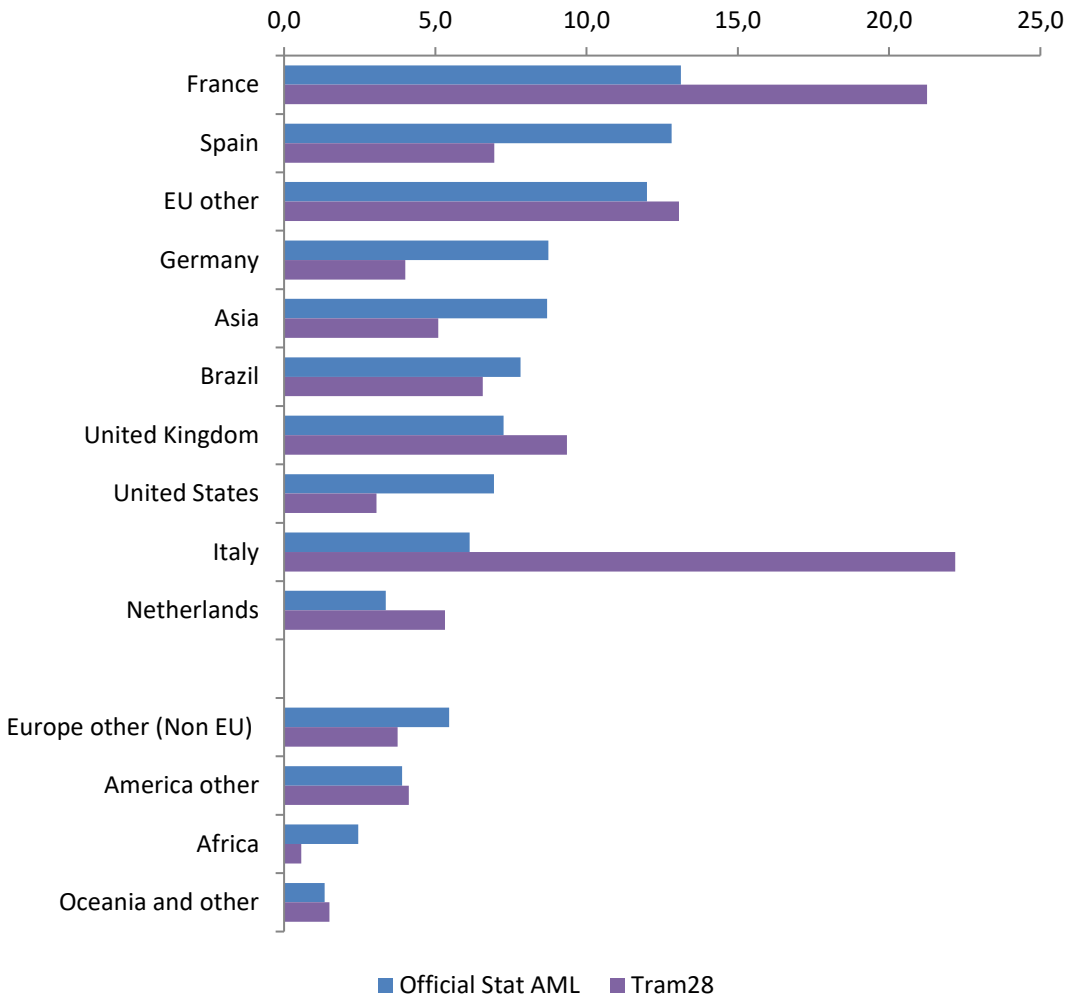


Country of residence: Official statistics and Jerónimos Monastery (TripAdvisor)



Data from Jerónimos Monastery shows us a similar structure we observed in Oceanário.

Country of residence: Official statistics and Tram 28 (TripAdvisor)



In this chart, Italy (again) and France appear with a predominance in TripAdvisor's data while Brazilian reviews have a significant percentage drop comparing to Oceanário and Jerónimos.

Tourism flows - Data by Month from TripAdvisor and Official Statistics

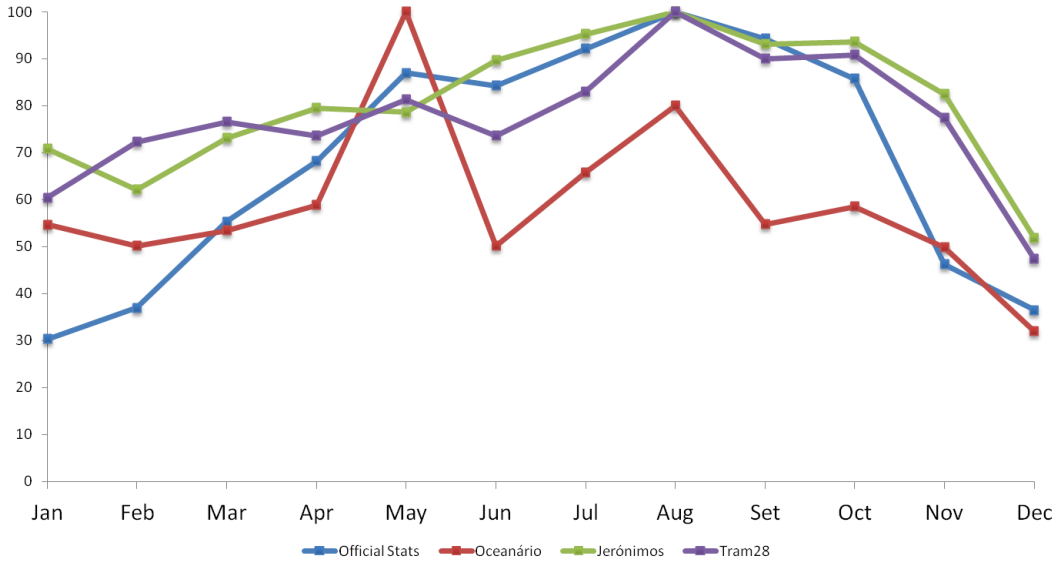
The reference official statistics for comparing with TripAdvisor's data are Non Resident Guests in hotel establishments by Geographic localization (Metropolitan Area of Lisbon, NUT II) and Type (hotel establishment); Monthly - Statistics Portugal, Guests stays and other data on hotel activity survey for 2016. Portuguese residents were excluded from this analysis.

For comparison purposes we used standardized values [4].

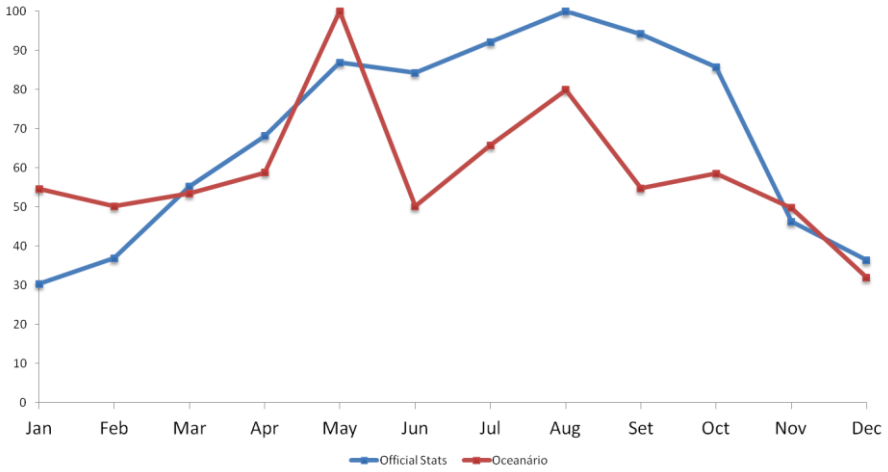
The purpose is therefore to compare trends and flows from both data sources.

[4] $(x \div max) \times 100$. Other forms of standardization were also tested, such as zscores and percentages, with similar results.

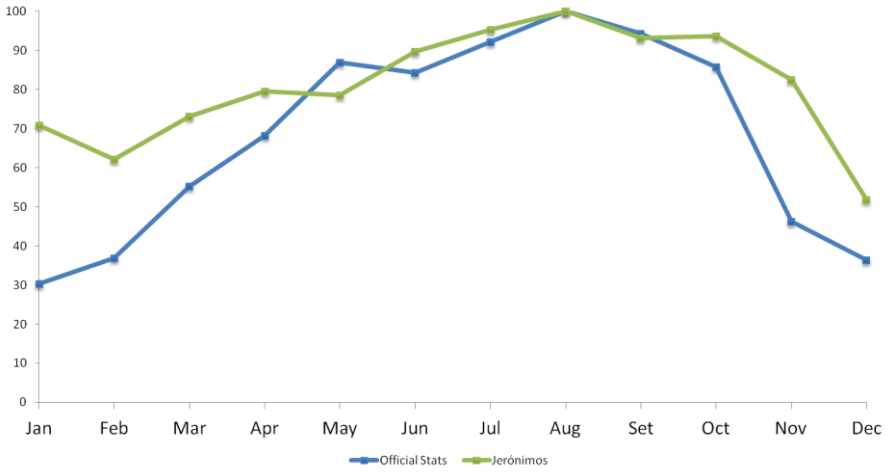
Monthly Guest stays (Official Statistics) and Monthly Reviews TripAdvisor



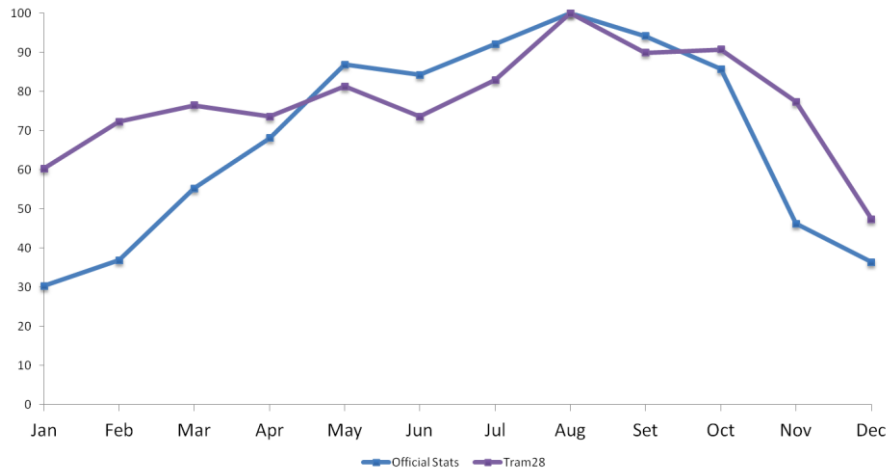
Monthly guest stays (Official Statistics) and Monthly reviews for Oceanário (TripAdvisor)



Monthly guest stays (Official Statistics) and Monthly reviews for Jerónimos Monastery (TripAdvisor)



Monthly guest stays (Official Statistics) and Monthly reviews for Tram 28 (TripAdvisor)



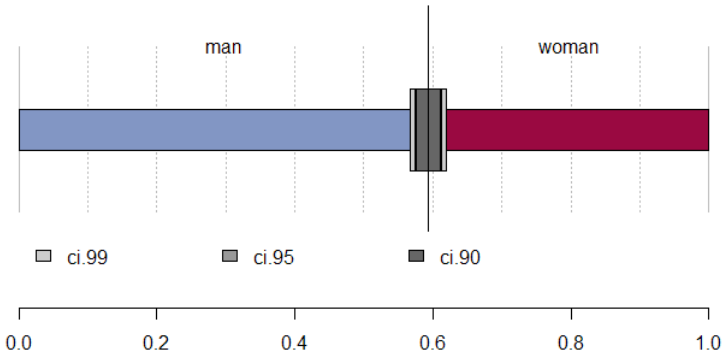
Data from TripAdvisor, particularly concerning Jerónimos Monastery and Tram 28, are remarkably similar in trend to official statistics. Both these tourist attractions diverge from official statistics trend in two particular moments: April/May and September/October. On the other hand, Oceanário has some particularities that at this time we are not able to interpret. Longer time-series and data for more tourist attractions reviewed in TripAdvisor would definitely be a valuable addition to this analysis. The differences in trend observed in these three tourist attractions might indicate different preferences among tourists from various countries of residence.

Oceanário Data by Gender

We were only able to extract gender from 42% users' reviews

Gender	Freq	Perc
Man	1401	59,3
Women	960	40,7
	2361	100

Length = 5624, NA = 3263 (58%)

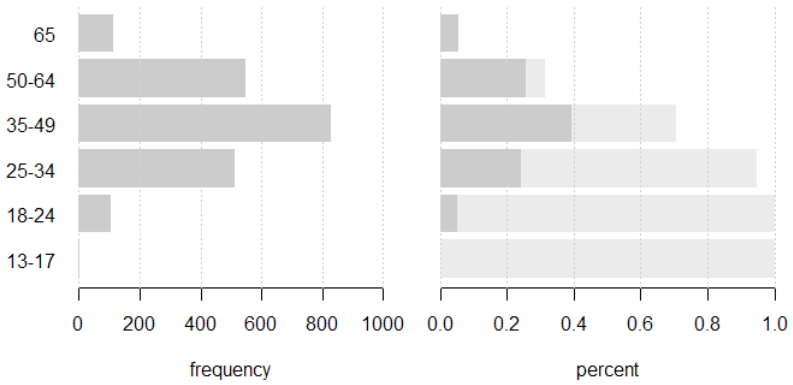


Oceanário Data by Age Group

We were only able to extract age group from 37,5% users' reviews

Age Group	Freq	Perc
60+	114	5,4
50-64	546	25,9
35-49	828	39,2
25-34	514	24,3
18-24	107	5,1
13-17	2	0,1
	2111	100

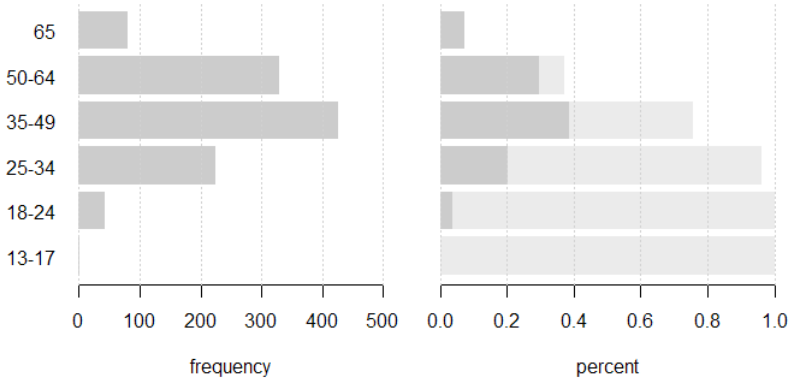
Length = 5624, NA = 3513 (62,5%)



Oceanário Data by Gender – Male and Age Group

Man by Age Group	Freq	Perc
60+	81	7,3
50-64	329	29,8
35-49	426	38,6
25-34	224	20,3
18-24	43	3,9
13-17	1	0,1
	1104	100

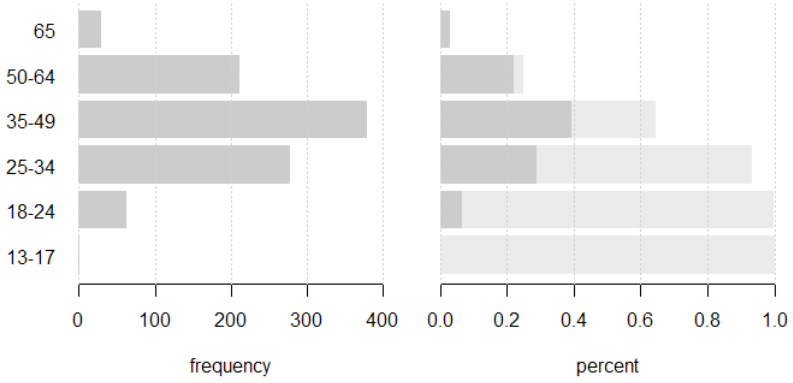
Length = 1401, NA = 297 (21,2%)



Oceanário Data by Gender – Women and Age Group

Women by Age Group	Freq	Perc
60+	30	3,1
50-64	211	22,0
35-49	378	39,4
25-34	277	28,9
18-24	63	6,6
13-17	1	0,1
	960	100

Length = 960, NA = 0 (0,%)



Final remarks and lessons learned

The TripAdvisor with more than 500 million traveler reviews is an extensive data source on tourism.

Web scraping has proven to be very efficient to extract data from the web and is a good alternative when there are no other sources; once the scripts have been developed the information can be extracted at any time.

The qualitative analysis of the collected reviews could not be done in due time, since the necessary and specific knowledge has been gradually acquired. However, it is a matter that will be part of our future work, given its importance and potential for application in several areas within NSI, such as COICOP classification, ISCO and NACE coding, etc.

Pre-processing tasks are time-consuming when working with data extracted from the web, therefore one should carefully plan this phase.

Since we don't have any knowledge about TripAdvisor user's behavior (eg.: if it differs from nationality), future work should include an extensive study of users profile.

Contextual information not included in TripAdvisor's data, such as significant events that could change the normal inflow of tourists from specific countries (holydays, religious events, music festivals, etc…) should be carefully addressed in future work as it may give some explanation for unexpected results.

Longer time-series and data for more tourist attractions reviewed in TripAdvisor would definitely be a valuable addition to this analysis.

Finally this work should be considered as exploratory and as an excellent opportunity to test and learn new tools that are of great importance to future work in NSI.

Future perspectives: description of the work package 7 – Portuguese Team

Population Domain

Aim and description:

- **Data sources:** Web portals (Ex. Tourism related portals) and traditional Tourism surveys
- **Methodology:** Webscraping, Data/Text/Web mining (eventually Machine learning). Quantitative and qualitative analysis of reviews / comments. Comparative analysis with traditional statistics.
- **The goal of the case study:**

Extract reviews and comments posted in internet portals by tourists and their profile or characteristics / attributes such as location, gender or other.

  - Compare results with traditional tourism surveys (tourism flows and visitors nationalities)

- **Main benefits and value added for official statistics:** provide timely insights of phenomena that are difficult to measure through traditional surveys.

Work plan for SGA-2

- List several potentially interesting tourism web portals

- Select the best source and justify the choice
- Determine a time frame for all data (for both web and traditional sources)
- Detailed description of relevant information found on the selected web portal (attributes, profiles, etc…)
- Select attributes / profiles to extract (justify choice)
- Select the appropriate tool for data extraction and pre-processing
  - Write the necessary scripts for extraction
  - Pre-process extracted data
- Select the database for storing the extracted data
- Select the data analysis tool
- Analyze data
- Final report will include:
  - Lessons learned
  - How future work could explore the results
  - Conditions for future use of these data sources within official statistics
  - Describe the added value

Appendix 2. Brief description of pilots

Below you can find a brief description of successful pilots. It is based on the WP7 Milestone 7.9 – List of possible pilots.

Life Satisfaction by Twitter

USE CASE NAME **Life Satisfaction by Twitter** (countries that can provide results as of February 2018: Poland, Portugal)

<table>
<tr>
<td>LEGAL ASPECTS
Twitter data may be collected via Twitter API. Personal data is limited to name (screen name), location.</td>
<td rowspan="3">TESTING/USEFUL REFERENCES
POLAND: The 6.187 tweets were used in implementation in Polish language, processed with stemming, lemmatization and stop words issues. Classification was made manually to prepare the training dataset, according to the list of classes presented below.

PORTUGAL: The 2.370 tweets were submitted to the following pre-process: Portuguese PorterStemmer; In CountVectorizer; Custom portuguese stopwords list (combined from several available Portuguese stopwords lists); Preserved accentuation with strip_accents = 'unicode'; Parameter max_features wasn’t subject to any restriction; the whole corpus was considered.</td>
</tr>
<tr>
<td>QUALITY (1-5)
availability – 5
sustainability – 4</td>
</tr>
<tr>
<td>METHODOLOGY
Machine learning – supervised learning
Web scraping – we use Twitter API to gather and process the data.

TECHNICAL REQUIREMENTS
Python
MongoDB
Apache Spark for efficient processing</td>
</tr>
<tr>
<td colspan="2">RESULTS / REMARKS / MAIN FINDINGS

Statistics Poland findings:
1. Due to the rules in Polish language, the and testing datas had to be prepared very precisely and pre-processed using ad-hoc stemming and lemmatization .
2. In the first phase of the implementation we had to increase the number of tweets used in the training dataset due to the problems with low accuracy in the results of the training dataset.
3. Increasing the number of tweets added a significant boost to accuracy.
4. A small training dataset and the disproportion on the number of cases in different classes (e.g., 70% of positive comments compared to 10% of negative) resulted in almost accidental classification of tweets.

Statistics Portugal findings:
Each sentiment should be accurately defined to avoid imprecision both in the extraction step as well in the manual validation step. Since they are not mutually exclusive some of the sentiments may have overlap meanings, e.g., happy/calm or discouraged/depressed. There are limitations to the classification when you don’t properly conceptualised what you are trying to measure.
1. These sentiments can be polarized e.g., happy/sad or discouraged/encouraged, but others cannot (depressed).</td>
</tr>
<tr>
<td colspan="2">IMPLEMENTATION
We decided to prepare training and testing datasets and used supervised learning in Python 3 language. INSEE Portugal modified Polish Software to work with Python 2.

The Training dataset for Portugal had 2.370 tweets (Portugal) and 6.187 tweets (Poland), (UK)for the 6 categories (1.happy, 2.neutral, 3.calm, 4.upset, 5.depressed and 6.discouraged).

In Portugal an additional twitter dataset with 600 tweets (100 for each category) was used for category prediction.
In Poland the testing dataset was 15% of the training population – it is a common way of providing information about the training dataset accuracy.</td>
</tr>
</table>

2. Specify the domain: We are measuring sentiments regarding to what? Sentiment expression on politics (e.g., EU Commission) or a social event (e.g., Olympic Games) rely on different vocabulary and therefore it should be used a different approach. If the domain is purposely vague and undetermined then the sample should be much larger.
3. Too many categories
4. Larger sample training dataset
5. Twitter is not the most popular social media platform in Portugal. This is not a problem in an experiment, but it is relevant issue for further developments.
6. As in opinions, we should consider 5 elements when classifying sentiments:
    a. Target entity/object: "I'm calm" VS "She's calm"
    b. Aspect/feature of the entity: "She has a calm voice" VS "She is calm"
    c. Sentiment value of the opinion from the opinion holder "She's barely calm" VS "She's completely calm"
    d. Opinion holder/opinion source: " I think she's calm" VS "She thinks she's calm"
    e. Time when the opinion is expressed: "I was depressed" VS "I am depressed"

SHORT DESCRIPTION

The goal of the use case is to deliver data on life satisfaction - 1.happy, 2.neutral, 3.calm, 4.upset, 5.depressed and 6.discouraged. The goal is to support the data from EU-SILC survey with more recent data. The major drawback from this case study is that the dataset may not be representative, which is an issue for the future pilots.

Life Satisfaction by Facebook

USE CASE NAME **Life Satisfaction by Facebook** (ONS UK)

LEGAL ASPECTS

Facebook comments to public news pages can be collected up to a week before from the Facebook Graph API. Personal information is limited to screen name

TESTING/USEFUL REFERENCES

Hundreds of thousands of comments were collected in the time frame of a month from a single news page. Positive/Negative sentiment was derived from comments comparing 4 different Sentiment lexicons and the Vader library

QUALITY (1-5)

availability – 4

sustainability – 4

METHODOLOGY

Sentiment lexicons used and score calculated based on sum of scores of individual words, averaged across sentences

Web scraping – we use Facebook API to gather and the data.

TECHNICAL REQUIREMENTS

Python

MongoDB

IMPLEMENTATION

We decided to compute the sentiment score (from -1 to 1) by extracting lexicon words from the comments and combining the individual word scores first by sentence, then for the whole comment.

Clerical review of comments sentiment

Approx. 1000 comments for the 3 categories (1. positive, 2. neutral, 3. negative)

RESULTS / REMARKS / MAIN FINDINGS

ONS findings:

There are several limitations that arose from using lexicon-based sentiment analysis methods for analysing Facebook comments. Some of them include:

1. Long text
2. Noisy comments: many comments with just a name in it
3. Context relevant
4. Keyword-based approach is totally based on the set of keywords. Sentences without any keyword would imply that they do not carry any sentiment at all.

5. Meanings of keywords could be multiple and vague, as most words could change their meanings according to different usages and contexts.
6. The Facebook API does not provide any other information than person's full name and ID. Additionally, users' IDs are meaningful only within the same API/app. This limits the understanding of representativeness issues
7. Case study was limited to a single news agency, whereas should include multiple of them to be as representative as possible
8. Using an emotion lexicon (where each word is rated according to 8 different emotional states: joy, sadness, fear, anger, disgust, surprise, anticipation, trust), rather than just positive/negative, didn't provide any robust finding

SHORT DESCRIPTION

The purpose of the case study was to examine the level of daily satisfaction by analyzing the content of messages for the presence of defined expressions describing emotional states, e.g., positive and negative sentiments and understand how they varies over time.

Tourism accommodation establishments

USE CASE NAME: **Tourism accommodation establishments** (Statistics Poland) (the result below shows issues related to boarding houses in one of the most popular touristic place in Poland – Kołobrzeg city)

LEGAL ASPECTS

Legal aspects are related to the possibilities of web scraping information on tourism accommodation establishments from websites such as booking.com, tripadvisor.com and related.

TESTING/USEFUL REFERENCES

We have tested using different data sources – web portals, including the most popular but also very specified oriented to tourism accommodation establishments (boarding houses) – booking.com, tripadvisor.com, agroturystyka.pl.

QUALITY (1-5)

availability - 5

sustainability - 3

METHODOLOGY

Web scraping – extracting data from tourism web portals and loading them into database (currently csv files).

Information from the Internet was used and compared with information from tourism register used for statistical survey.

TECHNICAL REQUIREMENTS

Python 3

IMPLEMENTATION

We have used the data from various Internet Portals to find tourism accommodation establishments (boarding houses).

The code was written in Python 3 and data was stored in CSV files.

RESULTS / REMARKS / MAIN FINDINGS

The following description shows a case of wrong classification in accommodation offering web portals, according to official data:

1. As part of the work, 17 objects that used the name of the boarding house (in Polish pensjonat – boarding house) were detected.
2. Based on web scraping, it is difficult to determine if a given object meets the mandatory requirements related to the official classification of an object for a given type of facility.
3. Based on the analysis carried out and a comparison of the list of objects obtained from scraping with the list of units in the tourist accommodation survey card, it was established that only one object submits a statistical report as a boarding house. Others are classified as objects of other types.
4. On the basis of direct contact with representatives of the city of Kołobrzeg having information on the number of tourist accommodation establishments classified as a boarding house, it

turned out that there is only one such facility that has obtained the appropriate status, meeting the requirements imposed by law.

5. The web scraping method is not applicable to searching for tourist accommodation facilities by type of facility. Therefore, we have to combine this data with official register of Tourism accommodation establishments

SHORT DESCRIPTION

The goal of the use case is to web scrape all information related to tourism accommodation establishments. The aim of the work described in remarks was to identify tourism accommodation establishments providing services as boarding houses in the town of Kołobrzeg.

Border movement

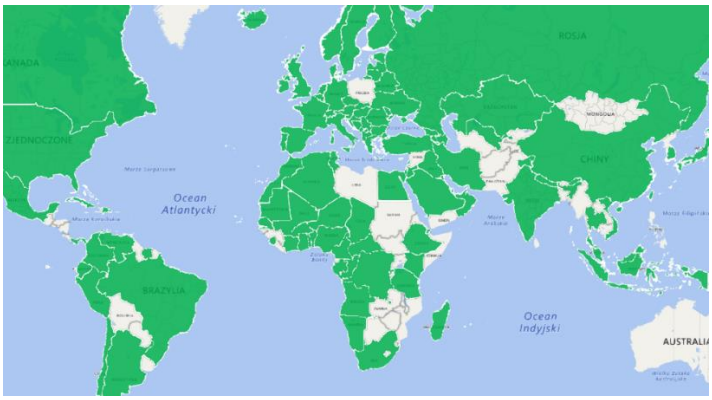
USE CASE NAME: **EU Internal Porter Crossing by air, maritime and land border traffic estimation** (Statistics Poland)

LEGAL ASPECTS	TESTING/USEFUL REFERENCES
Data from Polish Authorities, sample survey and websites with web scraping allowed. No legal obstacles.	We have tested machine generated data from road authorities and air traffic
QUALITY availability 4 sustainability 5	IMPLEMENTATION Obtained data concerns 29 airports: 15 Polish airports and 14 hub airports to 484 airports in 145 countries. It contains information on around 270 thousand flights. Aircraft technical information on number of seats with respect to class pertain to 105 types of aircrafts. Base of IATA, ICAO and FAA codes covers 10259 airports. Data from traffic sensors comes from countries like Poland, Germany and more listed below. Data on maritime traffic comes from government authorities.
METHODOLOGY Non-extensive Cross Entropy Econometrics, GLS and benchmarking	
TECHNICAL REQUIREMENTS Python, RStudio	

RESULTS / REMARKS / MAIN FINDINGS

1. Big Data on air traffic may increase countries coverage for which statistics on trips are produced. We obtained estimates for 145 countries while sample survey covers up to 80 countries.

   Graph. Countries with non-zero estimate from Big Data and sample survey.



2. Estimates are much more stable in time and distributions are smoother than in sample survey. For instance, with sample survey solely we reached only 6 destinations in Latin America and the distribution seems to be distorted and unreliable. At the same time with Big Data we reached 22 destinations out of around 30 possible destinations.
3. Big Data needs assistance of sample survey with respect to travel behaviours such as frequency of using hub airports, share of residents in total traffic etc.
4. Distribution of flights reflects preferences of many nations – not Polish directly. Therefore, obtained results may be more European than Polish

5. Data on road traffic intensity from traffic loops may partially replace manual counting.
6. With traffic loops only, we were able to cover 75.7% of land border traffic. Increasing density of traffic loop's net may improve the coverage.
7. Data sources and methods are completely different for air traffic and land border traffic.

Table. Border traffic with respect to country and dimension (land, sea and air) in 2015

	Number of crossings		
	Total	Polish residents	Non-residents
Total	217 175 084	88 121 949	129 053 135
land border	187 972 038	70 960 914	117 011 124
external EU's border	35 031 906	6 371 720	28 660 186
Russian-Polish border	6 098 860	3 369 138	2 729 722
Belarusian-Polish border	7 817 371	863 830	6 953 541
Ukrainian-Polish border	21 115 675	2 138 752	18 976 923
internal EU's border	152 940 132	64 589 194	88 350 938
Lithuanian-Polish border	4 946 268	1 528 605	3 417 663
Slovakian-Polish border	15 271 001	6 864 510	8 406 492
Czech-Polish border	50 870 006	24 917 817	25 952 189
German-Polish border	81 852 856	31 278 263	50 574 593
maritime border	1 851 524	1 088 040	763 485
air border	27 351 522	16 072 995	11 278 527
external EU's border	1 037 256	611 388	425 867
Russian-Polish border	315 118	185 740	129 378
Belarusian-Polish border	49 997	29 469	20 527
Ukrainian-Polish border	672 141	396 179	275 962
internal EU's border	3 947 888	2 326 998	1 620 890
Lithuanian-Polish border	90 604	53 405	37 199
Slovakian-Polish border	54 588	32 176	22 412
Czech-Polish border	179 694	105 917	73 777
German-Polish border	3 623 002	2 135 501	1 487 501
other air borders	22 366 379	13 134 608	9 231 770

SHORT DESCRIPTION

The aim of this study is to estimate air, maritime and land border traffic with respect to destination country in a case of air traffic.

For land border traffic data from traffic sensor in Poland, Germany, Slovakia and Lithuania was collected. Traffic sensors relevant to border traffic estimation were selected. For each source coefficient of variation was assigned basing on technical parameters of loops. Data biasedness was partially reduced with historical time series. Mirror statistics were combined with GLS method. Missing data was imputed basing on cross entropy measure. It enabled to estimate land border traffic with respect to country, vehicle and month.

For air traffic we gathered data on flights schedules from web scraping of flight movement webpages, data on technical information on aircrafts, base of IATA, ICAO and FAA codes, administrative data from Civil Aviation Authority of Poland, sample survey on tourism. Web scraping covered all Polish airports and crucial airports in Europe. Databases were linked into one database with country code (IACO or IATA), airport code and aircraft type as a key. Conditional distributions of passenger from each country were calculated and combined. Some statistics such as frequency of using hub airports, share of residents in total traffic were taken from sample survey on tourism. Data from Civil Aviation Authority of Poland were used to benchmark number of passengers to each hub country. Irrelevant routes were removed from database. It enabled to estimate number of passenger with respect to country using direct flights from Poland and using hub airports.

Crop types identification (1st methodological approach) by Statistics Poland

USE CASE NAME **Crop Types Identification (1)** using satellite imagery Sentinel-1 and Sentinel-2 for crops classification (Statistics Poland)

LEGAL ASPECTS

Sentinel data may be downloaded via ESA open access hub. Sentinel data are free and available for anyone.

TESTING/USEFUL REFERENCES

One-year time series raw satellite data for Poland.

Over 5.000 in situ samples were collected as a ground truth for satellite images classification.

QUALITY (1-5)

1. Accuracy – 4 (on average 80-97%)
2. Completeness – 5
3. Sustainability – 5
4. Availability – 5

Measured also by classification results compared to control in situ dataset (confusion matrix) and classification results compared to official administrative data (CSO, ARMA).

METHODOLOGY

1. Raw satellite image processing
2. Segmentation of processed data.
3. Object based image classification (machine learning algorithms).

TECHNICAL REQUIREMENTS

Workstation with high computing power (min. Core i7 processor, 64GB of RAM and 2TB SSD drive).

ESA SNAP and PolSARPro software for satellite data processing.

E-cognition Developer software for object-based image classification.

IMPLEMENTATION

- **Sentinel-1 A & B** From the beginning of the year 2017 more than 2200 scenes were downloaded for the whole country (3.75 TB). Data were downloaded in a fully automatic way. The procedure for image automatic downloading was developed in Earth Observation Group of Space Research Centre of Polish Academy of Sciences. Surface of Poland was covered by 6 paths of Sentinel-1 satellites.
- **Sentinel-2** Images were acquired in order to support the segmentation process and to improve winter and summer crops separation. More than 1160 images were downloaded, around 1.2 TB of data.
- **NDVI** Basing on a series of optical Sentinel-2 images from the period autumn 2016 and spring 2017 NDVI index was calculated from each image.
- **Segmentation** It was done in the eCognition software. In the first phase, LPIS vector was used for the exclusion of the areas that are outside of the arable lands (e.g., forests, water, built-up areas). First level of the segmentation was done based on the LPIS vector. Next, multitemporal radar images (lambda) were used for the creation of another level of segments. In this level, good separation of inhomogeneous arable lands (that occupy one field) was obtained.
- **Crop recognition** Based on the previous experience, the time series of the coherence matrix and H / L decomposition were classified using the Support Vector Machine (SVM) algorithm.

RESULTS / REMARKS / MAIN FINDINGS

Obtained results allow us to formulate following conclusions:

1. New method of in-situ samples gathering ensured their proper spatial distribution. This allowed proper learning of the classifier
2. Utilization of multi temporal NDVI mosaic obtained using data acquired between fall 2016 and spring 2017 allowed for more precise distinction of winter and summer crops.
3. Accuracy of classifier learning varies from 80% for path 3 to 97% for path 2. More detailed analysis of this phenomenon is needed. It means that the algorithms manages to correctly classify a crop from the images in 80% of the times.

Area [ha] of classified crops (preliminary results):

	winter wheat	corn	spring cereal mixes	winter rape	spring barley	oat	winter triticale	winter barley	winter cereal mixes	rye	spring wheat	spring triticale	spring rape
	2	3	4	5	6	7	8	9	10	11	12	13	14
Dolnośląskie	173253	145881	47073	154136	59687	41720	47594	39244	14211	42753	49929	11192	18243
Kujawsko-pomorskie	106582	221238	99912	136147	62746	37729	67247	56993	19979	90421	36283	19340	18053
Lubelskie	80226	134882	54770	104419	61947	54242	127243	16081	53798	87439	79212	68082	51356
Lubuskie	37762	53809	44258	52875	25441	21838	44464	19008	9486	32579	21665	12868	11471
Łódzkie	82661	127058	117794	49828	81216	86792	68448	36679	34115	87062	43037	19569	21436
Małopolskie	73606	55655	18170	30125	39699	14592	26300	21019	19546	14241	31585	7411	13414
Mazowieckie	128534	285548	93481	79358	106266	94890	115691	98596	61298	123378	72642	62513	28074
Opolskie	79370	60109	19870	66634	24564	11434	28393	20131	7224	23074	19036	5672	9106
Podkarpackie	43720	48561	12391	27083	21252	25955	37957	27630	13067	33414	43715	28357	40337
Podlaskie	57053	148827	72773	22294	44103	64786	59429	23586	31177	78835	51738	29961	12161
Pomorskie	94457	80850	62211	108989	49716	56912	65305	21061	16989	55091	49370	14028	9681
Śląskie	63784	42242	27681	31258	19671	21060	27327	9276	9928	18534	28313	7033	7325
Świętokrzyskie	71479	55883	28618	21855	44362	29265	23556	19767	10970	15941	39932	6231	9594
Warmińsko-mazurskie	160519	121593	62298	114711	48849	31884	95813	34177	29862	78641	74727	36332	17037
Wielkopolskie	130171	323677	149718	152008	120466	88570	135491	94627	27284	161912	50577	28547	29495
Zachodniopomorskie	90595	86818	66665	136496	56379	45216	64045	26356	11254	82772	78270	22716	26544

Confusion matrix for classified crops (1-st compilation):

Klasyfikacja \ Próbka	winter wheat	corn	spring cereal mixes	winter rape	spring barley	oat	winter triticale	winter barley	winter cereal mixes	rye	spring wheat	spring triticale	spring rape	Sum
winter wheat	27	0	0	0	0	0	1	0	1	2	1	0	0	32
corn	0	11	0	0	0	0	0	0	0	0	0	0	0	11
spring cereal mixes	0	0	17	0	0	0	0	0	0	0	0	0	0	17
winter rape	0	0	0	9	0	0	0	0	0	0	0	0	0	9
spring barley	0	0	0	0	20	0	0	1	0	0	0	0	0	21
oat	0	0	0	0	0	29	1	0	0	0	0	0	0	30
winter triticale	0	0	0	0	0	0	28	0	0	0	0	0	0	28

winter barley	0	0	0	0	0	0	0	12	0	0	0	0	0	12
winter cereal mixes	0	0	0	0	0	0	0	0	9	0	0	0	0	9
rye	0	0	0	0	0	0	0	0	0	24	0	0	0	24
spring wheat	0	0	0	0	0	0	0	0	0	0	17	0	0	17
spring triticale	0	0	0	0	0	0	0	0	0	0	0	12	0	12
spring rape	0	0	0	0	0	0	0	0	0	0	0	0	4	4
unclassified	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sum	27	11	17	9	20	29	30	13	10	26	18	12	4	
Producer	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.92	0.90	0.92	0.94	1.00	1.00	
User	0.84	1.00	1.00	1.00	0.95	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	
Hellden	0.92	1.00	1.00	1.00	0.98	0.98	0.97	0.96	0.95	0.96	0.97	1.00	1.00	
Short	0.84	1.00	1.00	1.00	0.95	0.97	0.93	0.92	0.90	0.92	0.94	1.00	1.00	
KIA Per Class	1.00	1.00	1.00	1.00	1.00	1.00	0.92	0.92	0.90	0.91	0.94	1.00	1.00	
Overall Accuracy	0.97													
KIA	0.97													

SHORT DESCRIPTION

The main goal of the use case is to deliver methodology for calculating area of main crops in Poland using free remote sensing data.

Crop Types Identification (2nd methodological approach) by CSO Ireland

USE CASE NAME **Crop Types Identification (2)** by the use of satellite imagery for crop classification (CSO Ireland)

LEGAL ASPECTS

Sentinel data may be downloaded via ESA open access hub. Sentinel data are free and available for anyone.

TESTING/USEFUL REFERENCES

Used a sample area of Irish satellite data.
LPIS data was used for training.
Data covered a sample time period.

QUALITY (1-5)

Availability – 5
Sustainability – 4

METHODOLOGY

Raw satellite image acquisition and processing
Preparation of LPIS data
Classification using various machine learning algorithms

TECHNICAL REQUIREMENTS:

Workstation with higher than average computing power (min. Core i7 processor, 8Gb RAM, powerful graphics card).
ESA SNAP software for satellite data processing.
QGIS and the Orfeo toolbox for image processing and classification.

IMPLEMENTATION

- Sentinel 1 and 2 images for May to July 2016 were downloaded for the whole of Ireland. Due to processing constraints a smaller area was used. This was selected based on variety of land cover types and minimizing cloud cover for the period of interest.
- LPIS data used was restricted to the same period and the same area to ensure the crops that would be in evidence in that time frame were being used in training.
- Classification was performed using the QGIS software using a variety of algorithms

RESULTS / REMARKS / MAIN FINDINGS

Conclusion:

1. The methodology did successfully classify data however the accuracy rate would need to be improved, with the limited sample and training set it was too low.

Issues:

1. Usable optical data was difficult to find for Ireland due to cloud cover.
2. Software was slow to pre-process, classify and post-process.
3. Spectral signatures of vegetative land cover types are quite similar which resulted in misclassified pixels.
4. Some classifiers often found Irish field sizes too small to generate spectral signature.

SHORT DESCRIPTION

The goal is to determine the feasibility and deliver the methodology for determining crop types in Ireland using free satellite imagery.

Agriculture and Tourism data combining

USE CASE NAME **Impact of landscape resources in rural areas on agritourism intensity** (Statistics Poland)

LEGAL ASPECTS

Data from registers, sample surveys and websites with web scraping allowed. No legal obstacles.

TESTING/USEFUL REFERENCES

None

QUALITY

availability 5

sustainability 5

METHODOLOGY

Spatial disaggregation methods, shrinkage estimator.

TECHNICAL REQUIREMENTS

Python, RStudio

IMPLEMENTATION

With web scraping we collected data from 1950 agritourist lodgings. Sample survey on homesteads covered over 1600 farmers while survey on tourism covered 150 records pertaining to agritourism: expenditures, nights spends etc. Land and Buildings Register is available on NUTS 5 level and contains detailed information on land structure and usage.

RESULTS / REMARKS / MAIN FINDINGS

1. Preliminary statistical analysis reveals that agritourist lodgings are mostly located on hilly areas. Number of them is moderately correlated with area of forest land, meadows, and pastures, while weakly correlated with area of arable land, lakes and rivers.
2. Number of agritourists is mostly related to area of lakes and rivers, then to forest land. It is negatively correlated with area of urban areas and arable areas.
3. With points 1-2 we may conclude that conditions of agritourism development are not the same as for agriculture.
4. Raw estimates on agritourism on NUTS 2 level are unreliable. It is ill-posed inference problem therefore OLS models seem to be irrelevant (mediocre R-squared while adjusted R-squared is low)

SHORT DESCRIPTION

The aim of this study is to disaggregate data on agritourism intensity from national level to NUTS 2 level.

Data collected in sample survey on agritourism does not allow to produce statistics on NUTS 2 level. We assumed that agritourist intensity is related to landscape resources of the region. Such a relation would help us to disaggregate data from national level to lower level of aggregation. We gathered data on landscape structure on NUTS 5 level from Land and Buildings Register from Central Office of Geodesy and Cartography, data on lowland / highland structure, data from sample survey on homesteads and survey on tourism on NUTS 2 level. Web scraping of webpages on agritourist objects was also performed to collect information on prices and locations.

The basic idea is to use spatial disaggregation methods to estimate statistics on NUTS 2 level. These methods consist of regression and benchmarking stage. Since we have very small amount of data on variables of interest, we may encounter ill-posed inference problem – raw estimates are unreliable. Therefore shrinkage estimators may be useful.